

Environmentally Friendly Organic Chemistry Laboratory Experiments for the Undergraduate Curriculum: A Literature Survey and Assessment

Ram S. Mohan* and Maria P. Mejia



Cite This: *J. Chem. Educ.* 2020, 97, 943–959



Read Online

ACCESS |



Metrics & More



Article Recommendations

ABSTRACT: Due to a growing awareness of environmental issues, green chemistry concepts are increasingly being incorporated into the undergraduate organic chemistry lecture and laboratory component. This minireview summarizes environmentally friendly organic chemistry experiments suitable for undergraduate laboratories. Whenever feasible, LD₅₀ values for various chemicals are provided to allow readers to determine the suitability of an experiment for their curriculum based on the toxicity of reagents used.

KEYWORDS: *Second-Year Undergraduate, Organic Chemistry, Safety/Hazards, Green Chemistry, Laboratory Instruction, Hands-On Learning/Manipulatives*

Nearly three decades have passed since the term “Green Chemistry” was coined and the now well-known ten principles of green chemistry were formulated.¹ Around the same time, the U.S. Congress passed the Pollution Prevention Act of 1990, which introduced the concept of source reduction.² Since then the field and idea of green chemistry has grown exponentially, and several seminal reviews have appeared on the subject. In 2001, Winterton introduced 12 more green chemistry principles that are particularly relevant to laboratory and research chemists.³ Recognizing that green principles are best introduced in the undergraduate curriculum, numerous educational initiatives have been made in the field of green chemistry. These include textbooks, summer schools, and workshops. Not surprisingly undergraduate organic chemistry curricula worldwide are undergoing revision to incorporate green chemistry concepts. The benefits of incorporating green and sustainable laboratory experiments are obvious: there is an increase in worker safety and reduction in cost associated with waste disposal. In addition, there is the less tangible but equally important benefit of inculcating students with lifelong sustainability principles. At least three classic books have highlighted contributions in the field.⁴

With a view of providing easy worldwide access to green organic chemistry undergraduate laboratory experiments, this review highlights several contributions in the field; however, in order to avoid excessive overlap with existing books, this minireview will focus on articles published since 2009 except when discussion of an earlier report is pertinent. Only experiments we consider suitable for undergraduate organic chemistry curricula are highlighted. While every effort has been made to be inclusive, any omission is inadvertent and not a judgment on a published article. Accompanying the enthusiasm for green and sustainable chemistry is often an overuse of the word “green” and exaggerated claims about greenness of a process. When appropriate this review will also point out the

less-green aspects of the experiments. This is not meant to be a criticism of the publication but is rather intended to help the reader make an informed decision about which laboratory experiments might be suitable for their needs. Ribeiro and co-workers have developed a metric to evaluate teaching laboratory experiments.⁵

Information about toxicity (LD₅₀ values) is given when available while acknowledging the drawback of extrapolating toxicity data from mice or rats to humans. The Hodge and Sterner scale is often used to evaluate the toxicity of reagents.⁶ To put things in perspective, it would be useful to point out to the reader that NaCl with an LD₅₀ of 3.0 g/kg (rat oral) is listed as slightly toxic while HgCl₂ with an LD₅₀ of 41 mg/kg (rat, oral) is listed as highly toxic. Anastas, Lasker, and co-workers have recently developed activities to introduce toxicology into the undergraduate chemistry laboratory.⁷ Such tools are necessary to allow chemists to design safer materials, and it is important to introduce this concept at the undergraduate level.

Safety in the laboratory has always been a primary motivation to make experiments safer. One of the early ways to make laboratory experiments greener and safer was by simply carrying out the reactions on a micro scale. Microscale reactions reduced cost of reagents and minimized the amount of waste generated. However, they were not necessarily green because the experiments did not always incorporate green chemistry principles. Additionally, microscale experiments often do not involve traditional organic synthesis techniques and hence do not always leave the student with a realistic picture of the techniques used in the industry. Microscale techniques also

Received: August 12, 2019

Revised: February 9, 2020

Published: February 21, 2020



ACS Publications

© 2020 American Chemical Society and
Division of Chemical Education, Inc.

943

<https://dx.doi.org/10.1021/acs.jchemeduc.9b00753>
J. Chem. Educ. 2020, 97, 943–959

generally lead to a lowering of yield, something certainly not desirable from a green chemistry perspective. Ribeiro and co-workers have developed a new risk index called SRI (Scale Risk Index) to assess improvements of safety upon reducing the scale.⁸ They argue that it is not possible to find a general correlation between green chemistry and microscale, and it is best to assess each specific reaction for its own merits.

As an increasing number of programs and departments incorporate green chemistry and seek curricular change, it is important to be aware of barriers to such change. Hutchinson has shown that using aspects of systems thinking makes it possible to develop greener organic chemistry laboratory curricula that are easier to adapt.⁹

The experiments summarized here are categorized by reaction type, though overlap between certain categories will be obvious to readers.

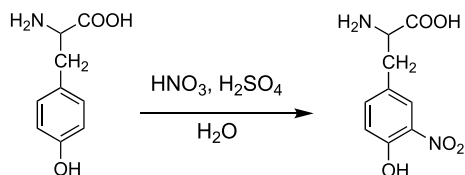
ELECTROPHILIC AROMATIC SUBSTITUTIONS

A significant amount of time is devoted in most undergraduate organic chemistry curricula to discussion of electrophilic aromatic substitutions (EAS). Some of the more important examples of EAS include Friedel–Crafts acylation and alkylation, halogenation, nitration, and sulfonation. Many EAS reactions require a Lewis acid to activate the electrophile, and this in turn leads to significant waste. Hence environmentally friendly methods to carry out electrophilic substitution reactions are of particular relevance to the undergraduate curricula.

Nitration of Phenols

Nitration of aromatics is one of the most important industrial examples of EAS because of the synthetic utility of nitro compounds. Nitration of aromatics is often used to teach the mechanism of electrophilic substitutions and many undergraduate laboratory curricula include nitration of aromatics as an experiment. The traditional method of nitration using a mixture of concentrated nitric and sulfuric acid is hazardous and generates considerable acidic waste. In addition, many nitro compounds are toxic and thus a toxic product is generated which must be appropriately disposed. Jones-Wilson and Burtch reported the nitration of *L*-tyrosine using a mixture of concentrated HNO₃ and concentrated H₂SO₄ to give 3-nitrotyrosine as a green experiment (Scheme 1).¹⁰

Scheme 1. Nitration of Tyrosine

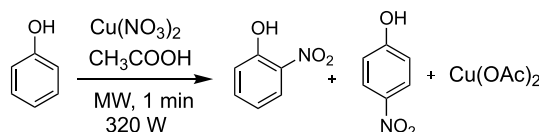


Unlike commonly used aromatic substrates for nitration such as acetanilide or phenol, tyrosine is relatively nontoxic (LD₅₀ = 5.1 g/kg, rat, oral). However, the reaction still suffers from large volumes of corrosive acid that must be neutralized, resulting in generation of significant waste. The authors do not report a yield, so it is hard to evaluate the efficiency of the reaction. In addition, the crude product is rinsed with ethyl acetate, leading to organic waste.

Ghalsasi and co-workers reported the nitration of phenol using Cu(NO₃)₂ under microwave irradiation as a green chemistry laboratory experiment (Scheme 2).¹¹ However, the

procedure has several drawbacks that do not make the method particularly green.

Scheme 2. Nitration of Phenol under Microwave Irradiation

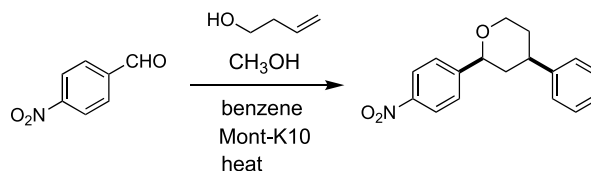


Phenol itself is quite toxic (LD₅₀ = 317 mg/kg, rat, oral) and great care must be exercised in handling it. The products are also toxic (*p*-nitrophenol, LD₅₀ = 230 mg/kg, rat, oral and *o*-nitrophenol, LD₅₀ = 334 mg/kg) and hence must be disposed of appropriately. The advantage of this method is that it avoids both nitric acid and sulfuric acid. However, copper nitrate is toxic (LD₅₀ = 940 mg/kg) and must be handled with care. A big drawback of this method is that product is isolated using silica gel chromatography, which generates considerable waste. In addition, silica dust is harmful and must be handled in a hood with a mask. Alternatively, the products can be isolated by cumbersome steam-distillation, which also generates significant acid waste. The products were analyzed using TLC (ethyl acetate/hexane). Although *n*-hexane is a known neurotoxin,¹² numerous academic laboratories continue to use *n*-hexane when a simple switch to heptane would result in a significant safety improvement. The authors report the procedure using both a commercial microwave (MW) oven and a domestic MW oven. Since it is very difficult to control temperature using a domestic MW oven, for safety reasons it is advised that they not be used for reactions. In fact, most major journals, including this *Journal*, no longer accept manuscripts that report reactions run using a kitchen MW oven.¹³

Prins–Friedel–Crafts-Type Reaction

Dintzner and co-workers reported a Montmorillonite K10 catalyzed synthesis of tetrahydropyrans (Scheme 3) via a Prins

Scheme 3. Synthesis of Tetrahydropyrans via Prins Reaction Followed by a Friedel–Crafts Reaction



reaction followed by Friedel–Crafts reaction.¹⁴ This experiment highlights several green chemistry principles such as use of multicomponent reactions,¹⁵ and a safer catalyst.

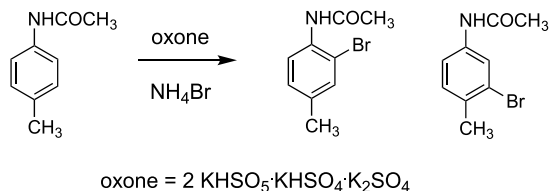
The product is isolated by filtration and no aqueous waste is generated. The major drawback of this method is the use of benzene, which is of course highly toxic and hence best avoided. It is unclear if the authors tried to replace benzene with toluene though the latter may likely give a mixture of products. One attractive feature of this experiment is that it is designed as an open-ended research-based project.

Bromination

Along with nitration, bromination is a common experiment frequently incorporated into the undergraduate laboratory curriculum to illustrate aromatic electrophilic substitution. One major difficulty with bromination is that traditionally it is

carried out with molecular Br_2 , a highly corrosive and hazardous reagent. An attractive alternative to Br_2 is offered by Hess and co-workers who reported the bromination of 4-methylacetanilide using oxone ($\text{LD}_{50} = 2.0 \text{ g/kg}$, rat, oral) and ammonium bromide in an aqueous solvent (Scheme 4).¹⁶ The substrate 4-

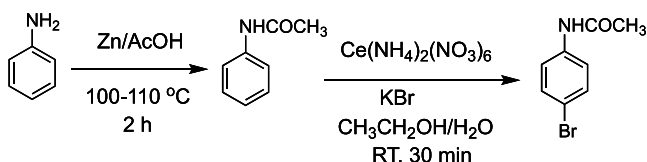
Scheme 4. Bromination of 4-Methylacetanilide Using Oxone



methylacetanilide itself is generated by acetylation of 4-methylaniline, which is highly toxic and hence adequate care must be exercised while handling it. The experiment is designed to introduce students to 1-dimensional selective nuclear Overhauser effect (NOE) NMR spectroscopy to determine the regioselectivity of the bromination reaction. Such advanced analysis provides students with a real-life experience of solving structure determination problems.

While in the previous method acetylation is carried out with corrosive acetic anhydride ($\text{LD}_{50} = 1.7 \text{ g/kg}$, rat, oral), Mukherjee and co-workers offer an attractive alternative method that is greener.¹⁷ They reported the acetylation of aniline using Zn dust and acetic acid ($\text{LD}_{50} = 3.3 \text{ g/kg}$, rat, oral). The zinc dust is believed to generate zinc acetate, which is stated to be more electrophilic than acetic anhydride. The resulting acetanilide is brominated using ceric ammonium nitrate (CAN, $\text{LD}_{50} = 300\text{--}3000 \text{ mg/kg}$, rat, oral) and potassium bromide (Scheme 5), and the products are isolated by pouring the reaction mixture onto ice.

Scheme 5. Synthesis of 4-Bromoacetanilide from Aniline



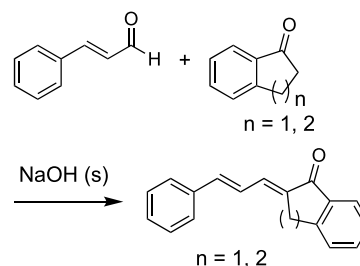
CARBONYL CONDENSATIONS

Various carbonyl condensations such as aldol condensation, Knoevenagel and Michael reactions are significantly emphasized in most year-long organic chemistry courses. Hence it is of interest to have accompanying laboratory experiments that illustrate that these reactions can be carried out in an environmentally friendly manner.

Aldol Condensation

Several experiments illustrating green aldol reactions have been reported.¹⁸ Jaworek-Lopes and co-workers have reported a crossed aldol condensation reaction between *trans*-cinnamaldehyde and a variety of ketones.¹⁹ The reaction is carried out in a vial under solvent-free conditions, and the product is isolated after acidification by filtration (Scheme 6). The use of relatively nontoxic substrates, and solvents for purification via recrystallization (ethanol or methanol) make this procedure relatively green. However, the method uses fairly standard conditions for aldol condensations and does not offer any new green reagents.

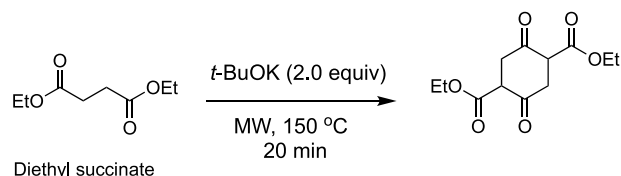
Scheme 6. Aldol Condensation between *trans*-Cinnamaldehyde and Ketones



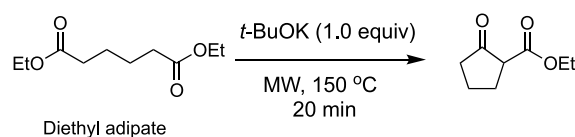
One nice feature of this experiment is that the extended conjugation in the product resulting from use of *trans*-cinnamaldehyde also allows for product analysis by UV spectroscopy, still a relatively uncommon spectroscopic technique in undergraduate organic chemistry laboratories.

Horta has reported microwave-assisted Claisen condensation (Scheme 7) and Dieckmann condensation (Scheme 8) experiments that are especially well suited for large laboratory sections.²⁰

Scheme 7. Microwave-Assisted Claisen Condensation



Scheme 8. Microwave-Assisted Dieckmann Condensation

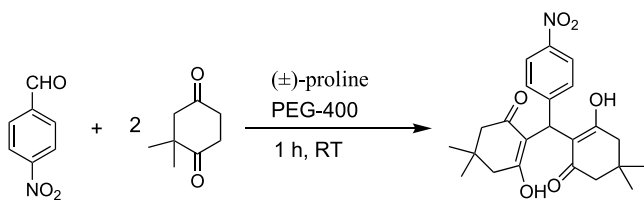


Although the authors state that the reactions can be carried out in a solvent-free fashion, they do use a high boiling solvent to better homogenize the reaction mixture that otherwise becomes a paste. Such high boiling ethers (1,4-dioxane or 1,2-diethoxyethane) are hazardous, as is DMSO. The use of a microwave oven with a carousel that can hold 40 reaction vessels makes this experiment well-suited for large undergraduate laboratory sections. The authors do offer caution concerning possible overheating in a microwave and suggest that reactions be visually monitored for any fume or smoke. Such caution must be exercised with any reaction carried out in a microwave oven as rapid heating can often lead to a run-away reaction.

Knoevenagel and Michael Reaction

Dicks, Nigam, and co-workers reported a semimicroscale green carbonyl condensation that demonstrates solvent and catalyst recycling.²¹ This experiment is carried out in polyethylene glycol (PEG-400) as a solvent. PEG solvents are relatively nontoxic, nonvolatile, and easy recyclable.²² In this experiment, which involves a Knoevenagel condensation followed by a Michael condensation, both the solvent, PEG-400, and the catalyst, ($\pm$)-proline, are recycled (Scheme 9). The experiment also allows students to realize that recycling solvents and catalysts comes at a cost, and this is something they must keep in mind when developing sustainable processes. The experiment could

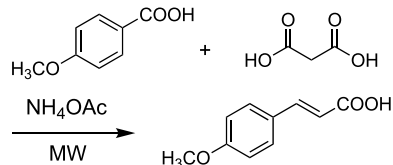
Scheme 9. Knoevenagel and Michael Reaction in PEG-400



also be adapted as a discovery lab wherein students are asked to elucidate the structure of the product.

Kumar and co-workers reported a facile and benign synthesis of cinnamic acids from aromatic aldehydes and malonic acid using ammonium acetate ($LD_{50} = 6.0$ g/kg, rat, oral) as a catalyst under solvent-free conditions.²³ The reaction was further facilitated by microwave irradiation. Inspired by this report, Keusemman and Morrow reported a green synthesis of *trans*-4-methoxycinnamic acid using ammonium acetate as a catalyst under microwave conditions (Scheme 10).²⁴ Unfortunately, the

Scheme 10. Microwave Assisted Synthesis of Cinnamic Acids



reaction was carried out in a domestic microwave oven. To minimize hazards associated with the use of a domestic microwave, the authors chose high boiling compounds and an open reaction vessel. Additionally, the reaction mixture was placed in a rather large bed of alumina powder. Although these precautions alleviate some of the concerns associated with use of a domestic microwave, such procedures should be used with utmost caution in a teaching laboratory in view of safety, as previously mentioned.

Mullins and co-workers reported an unusual Knoevenagel condensation to synthesize a carbocyclic curcuminoid (Scheme 11).²⁵ The experiment is particularly attractive for multiple reasons: it is a rare example of a mild way to use the less acidic hydrogen in a dicarbonyl compound. This is achieved using a boron enolate. The reaction proceeds via an electrophilic iminium ion generated in situ from the aldehyde and morpholine.

Additionally, the reaction is promoted using microwave irradiation. Several important chemical concepts such as acid–base properties, regioselectivity, and Lewis-acid-promoted condensation are illustrated in this experiment.

■ COUPLING REACTIONS

The last two decades have seen an explosion in the number and scope of coupling reactions, which have become the cornerstone

of many total synthesis efforts. The importance of coupling reactions received the ultimate validation in the form of the Nobel Prize being awarded to Heck, Negishi, and Suzuki in 2010 for developing palladium-catalyzed cross-coupling reactions. Historically, coupling was introduced in the undergraduate organic chemistry laboratory curriculum as an unwanted side reaction that occurred during a Grignard synthesis. For example, in the formation of phenylmagnesium bromide from bromobenzene, biphenyl is often formed as an undesirable coupling product. A modern-day version of such a coupling using a palladium catalyst is known as the Kumada coupling. It is only in the past decade that discussion of coupling reactions has made its way into standard organic texts. There are not many examples of coupling reactions suitable for undergraduate organic chemistry curricula. Hence, examples of green coupling reactions are of particular interest. Although coupling reactions involving palladium have received considerable attention from the synthetic community, the toxicity of palladium is often overlooked or not discussed. Toxicity is clearly not based on a single parameter and can depend on the oxidation state of the metal and the associated anion. Ananikov and co-workers have published an excellent discussion of the toxicity of metal compounds that dispels several long-held myths about toxicity.²⁶

Suzuki Coupling

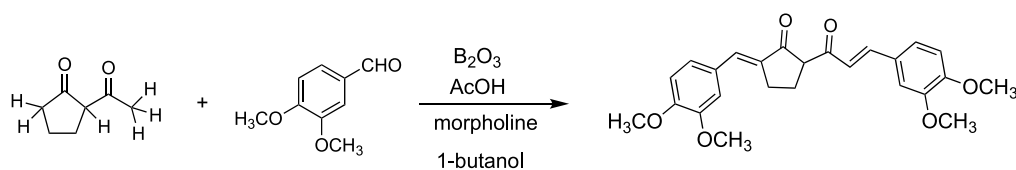
Rich and Pantess have reported a microscale $Pd(OAc)_2$ -catalyzed Suzuki coupling under aqueous conditions using TBAB (*tetra-*n*-butylammonium bromide*) that is suitable for undergraduate organic chemistry laboratories.²⁷ Students synthesize 4-bromoacetanilide and convert it to 4'-phenylacetanilide via a Suzuki coupling, thus minimizing waste generation (Scheme 12).

Deveau and co-workers reported a Suzuki coupling reaction under aqueous reactions that is used to synthesize ethyl (4-phenylphenyl)acetate, a precursor to Felbinac.²⁸ Felbinac is a topical medicine used to treat inflammation and arthritis. The target is synthesized using three different conditions (Scheme 13). The discovery-oriented nature of the experimental design allows students to determine relative efficiency of each method by considering overall atom economy (AE) and costs. The workup for each method involves extraction with diethyl ether, which makes the procedure not so environment-friendly. The authors mention that the ionic liquid, [bmim] PF_6 can be recycled, in principle. Caution must be exercised in doing so because the PF_6 anion is known to hydrolyze and generate highly toxic HF when heated, and in this experiment the ionic liquid is mixed with water.²⁹

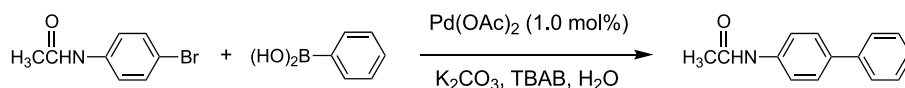
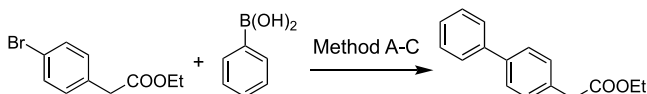
Borges and co-workers reported a microwave-assisted Suzuki coupling reaction to synthesize 5-phenyl-2-hydroxyacetophenone derivatives (Scheme 14).³⁰

The green aspect of this experiment comes from the use of a microwave oven to promote the reaction, and the low catalyst loading. The workup unfortunately utilizes CH_2Cl_2 . Refresh-

Scheme 11. Synthesis of a Carbocyclic Curcuminoid Using a Boron Enolate



Scheme 12. Suzuki Coupling under Aqueous Conditions

Scheme 13. Synthesis of Felbinac Using a Suzuki Coupling^a

^aMethod A: Pd(OAc)₂ (15 mol %), TBAB, K₂CO₃/H₂O, 90–95 °C, 1 h. Method B: Pd(OAc)₂ (1.8 mol %), Na₂CO₃, Acetone/H₂O, 40–45 °C, 1 h. Method C: Pd(OAc)₂ (0.5 mol %), Na₂CO₃, [bmim]PF₆/H₂O, 100–105 °C, 1 h.

ingly, however, the chromatographic purification uses heptane rather than the more common but toxic hexane.

Suzuki–Miyaura Coupling

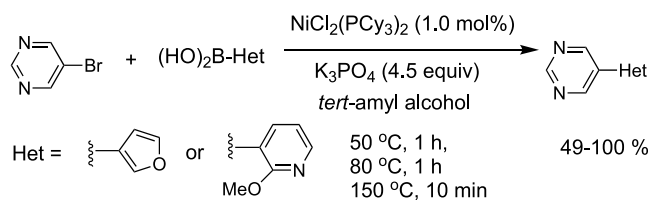
Garg and co-workers reported a nickel-catalyzed Suzuki–Miyaura coupling in two greener solvents, 2-methyl THF and *tert*-amyl alcohol.³¹ They utilized an air-stable commercially available precatalyst, bis(tricyclohexylphosphine)nickel(II) chloride, NiCl₂(PCy₃)₂. A modified version of this reaction (Scheme 15) was adapted as a green laboratory experiment by Garg and co-workers.³² Students are exposed to green chemistry concepts and to the importance of heterocycles in medicinal chemistry. Additionally, the replacement of palladium with nickel allows for discussion of metal residues in drugs and the need to use safer metals for drug synthesis in the pharmaceutical industry. Although the nickel catalyst is somewhat expensive (\$42/gram, Sigma-Aldrich), the experiment is carried out on a small scale, and remarkably enough only 4.4 mg of the catalyst is used for every 100 mg of bromopyrimidine, thus making the experiment suitable even for large laboratory sections!

Wissinger and co-workers reported a two-step synthesis involving iodination followed by a Suzuki–Miyaura coupling.³³ This experimental sequence adapts previously reported procedures for iodinations³⁴ as well as the Suzuki–Miyaura coupling suitable for undergraduate laboratories. It offers a greener alternative to electrophilic iodination, which is carried out using oxone and KI instead of molecular I₂, and the subsequent Suzuki–Miyaura coupling is carried out under aqueous conditions (Scheme 16).

Ballard has reported a green reductive homocoupling of bromobenzene to yield biphenyl (Scheme 17).³⁵ The method uses a relatively benign iron catalyst, iron(III) acetylacetonate Fe(acac)₃ (LD₅₀ = 1.87 g/kg, rat, oral), instead of more expensive and toxic palladium-based catalysts.

The experiment also serves to illustrate carbon oxidation–reduction concept, something undergraduate students recognize less easily than oxidation–reduction of metals. Since biphenyl is a side-product in standard Grignard additions of phenylmagnesium bromide to aldehydes, ketones, and esters, this

Scheme 15. Nickel-Catalyzed Suzuki–Miyaura Coupling



experiment could serve as a complementary laboratory experiment to illustrate that an undesired product in one reaction is sometimes the desired target in another.

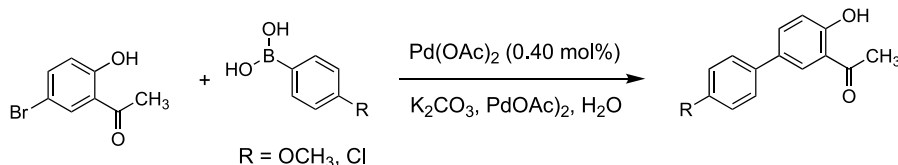
■ DIELS–ALDER REACTIONS

Nearly 50 years before the term green chemistry was coined, Otto Diels and Kurt Alder developed the Nobel Prize-winning Diels–Alder reaction, which to date remains one of the best examples of an atom-economical reaction, an important green chemistry concept. The authors described their findings in a series of 28 seminal articles published in *Justus Liebigs Annalen der Chemie* and *Berichte der deutschen chemischen Gesellschaft*. The reaction has been the subject of numerous reviews and books³⁶ and has also been studied in water as a solvent.³⁷

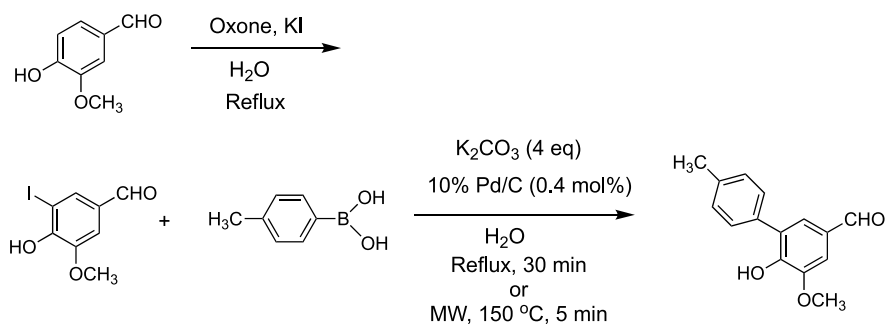
The Diels–Alder reaction is discussed extensively in most undergraduate organic chemistry curricula, and several laboratory experiments feature the Diels–Alder (D–A) reaction. The D–A is especially well suited to illustrate green chemistry concepts. Although the reaction of a diene and dienophile to give the cycloadduct is 100% atom economical, there are many instances when this is not the case, and such examples can be used to illustrate the limitations of atom economy to evaluate greenness of a reaction. For example, when the diene is generated in situ (1,3-butadiene, for use in a second-year laboratory is typically generated by thermal decomposition of 3-sulfolene) the reaction has an atom economy <100%. D–A reactions in undergraduate chemistry laboratories are also often run in a solvent. Solvents often end up in a waste jar but of course do not figure into the atom economy calculation, and this point can be used to point out to the student the limitations of using atom economy as a metric for efficiency. Rugutt and co-workers reported a D–A reaction between maleic anhydride and a variety of dienes in THF or *o*-xylene.³⁸ They describe the method as green but besides the inherent efficiency of a D–A reaction, it is unclear what makes this procedure particularly green, especially since the reactions are run in a solvent.

Amin and co-workers reported a D–A reaction between anthracene and maleic anhydride (LD₅₀ = 400 mg/kg, rat, oral) using a solar irradiation heat source (Scheme 18).³⁹ Although

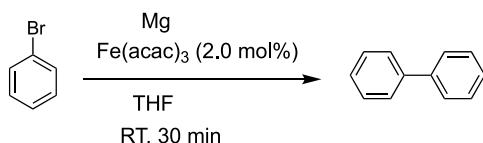
Scheme 14. Synthesis of Phenyl-2-hydroxyacetophenone Derivatives via a MW-Assisted Suzuki Coupling



Scheme 16. Synthesis of an Aryl Iodide Followed by a Suzuki–Miyaura Coupling

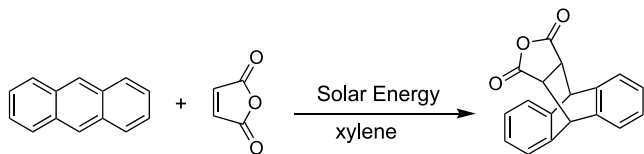


Scheme 17. Iron(III) Acetylacetonate Catalyzed Reductive Homocoupling of Bromobenzene

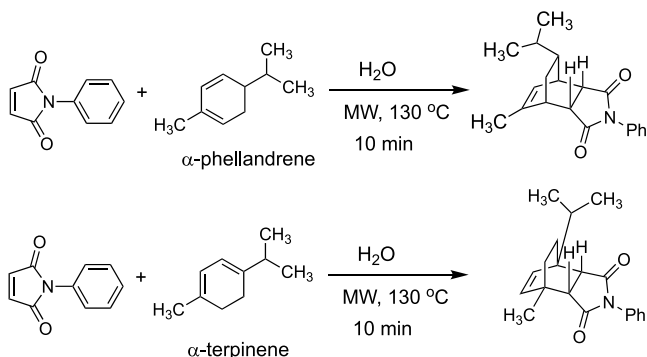


the reagents used are not particularly benign, the method does illustrate the green chemistry concept of alternative energy sources to promote reactions.

Scheme 18. Diels–Alder Reaction Promoted by Solar Energy



A popular discovery-based laboratory experiment involves identification of the major diene in eucalyptus oil based on the D–A product between this oil and maleic anhydride.⁴⁰ Although an interesting experiment, the procedure presents several safety concerns. The reaction is carried out in diethyl ether (LD_{50} = 1,211 mg/kg, oral, rat), which is highly flammable and is a neurotoxin. The concentration of phellandrene (the unknown diene) in eucalyptus oil is low and thus a large amount of the oil (2–3 mL) is required. Aslanian has reported a modified version of this method that alleviates some of the safety concerns.⁴¹ The reaction is carried out in water using a microwave as the heat source (Scheme 19). The experiment is carried out with

Scheme 19. Diels–Alder Reaction between *N*-Phenylmaleimide and Diene from Eucalyptus Oil

eucalyptus oil and the results are compared to reactions run using an authentic sample of α -phellandrene and α -terpinene. Since maleic anhydride used in the original procedure is moisture sensitive, it was replaced with *N*-phenylmaleimide (LD_{50} = 128–148 mg/kg). Unfortunately, this replacement comes at a cost as *N*-phenylmaleimide is considerably more toxic than maleic anhydride. However, this fact can also be used to illustrate to the student how complex evaluating the greenness of a reaction is.

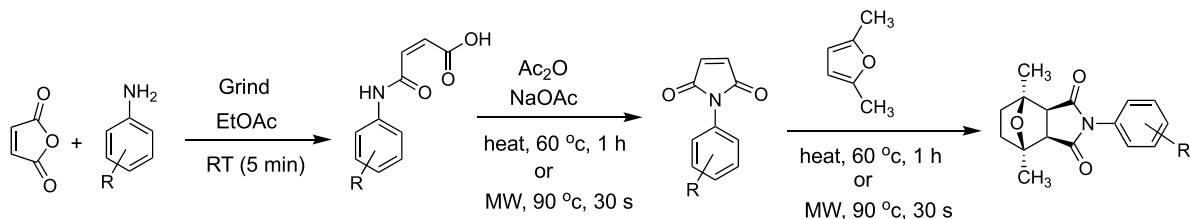
Bastin, Nigam, and co-workers reported a synthesis of substituted *N*-phenylmaleimides and demonstrated their use in a D–A reaction.⁴² Substituted *N*-phenylmaleimides are useful but expensive intermediates in organic synthesis. The multistep synthesis experiment illustrates several green chemistry principles, such as a solvent-free reaction in a mortar or use of a green solvent, ethyl acetate, and use of a microwave as a heat source (Scheme 20).

■ SYNTHESIS OF ACID DERIVATIVES (ESTERS AND AMIDES)

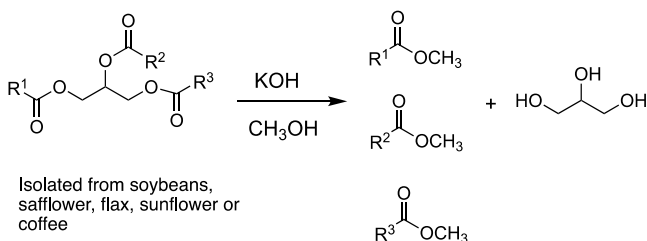
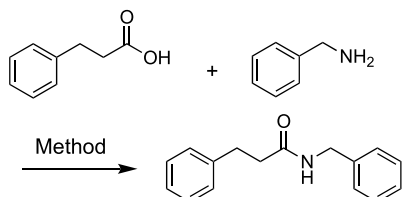
Acid derivatives are important functional groups that are discussed in almost all organic chemistry courses. Their chemistry, which allows students to see the interconnection between functional groups, often prominently features in multistep organic synthesis in the undergraduate curriculum. Experiments that illustrate synthesis of acid derivatives using green chemistry principles are increasingly common and desirable. Goldstein has reported a two-week experimental sequence wherein students isolate a triglyceride from a variety of nuts in the first week, and during the second week they convert the triglyceride into biodiesel (Scheme 21).⁴³ The experiment illustrates the important green chemistry concept that non-petroleum-based resources can be a valuable source of chemicals.

Amide bond formation is one of the most important reactions in medicinal chemistry and yet, ironically, it is among the less green reactions known. Hence, developing greener ways to form an amide bond is a priority in the pharmaceutical industry.⁴⁴ Fennie and Roth reported an experiment that illustrates amide bond formation whose efficiency is analyzed using green chemistry metrics.⁴⁵ The experiment involves an amide bond formation between hydrocinnamic acid and benzyl amine under three different conditions: (i) using an acid chloride to activate the carboxylic acid; (ii) using a coupling reagent (HATU); and (iii) condensation catalyzed by boric acid (Scheme 22).

Several green metrics such as isolated yield, atom economy, reaction mass efficiency, process mass efficiency, and ecoscale were used to determine the relative efficiency of each method. Such analyses enhance student understanding of green

Scheme 20. Multistep Synthesis of substituted *N*-Phenylmaleimides

Scheme 21. Synthesis of Biodiesel from a Variety of Nuts

Scheme 22. Comparison of Various Amide-Bond Formation Methods^a

^aMethod 1: (i) $(\text{COCl})_2$ (2.0 equiv), DMF (cat), CH_2Cl_2 , 0 °C, rt; (ii) concentrate; (iii) Add PhCH_2NH_2 (2.0 equiv), 0 °C, rt. Method 2: (i) HATU (1.2 equiv), $i\text{Pr}_3\text{NEt}$ (1.4 equiv), DMF, 0 °C, rt. Method 3: Boric acid (cat), Toluene, heat, Dean–Stark.

chemistry principles and drive home the point that determining efficiency is not as simple as picking between two parameters.

Diner and co-workers reported a synthesis of Lidocaine that is student driven and allows flexibility in the degree of independence instructors wish students in their laboratory sections to have.⁴⁶ The protocol allows students to study the effect of several parameters such as reaction temperature, equivalence of reagents, and solvent effects (Scheme 23). The use of excess starting material makes this procedure less green than claimed.

■ CONTINUOUS FLOW TECHNIQUES

Although pharmaceutical synthesis is still largely carried out in batch reactors, continuous flow is becoming an increasingly popular technique. It is especially advantageous when reactions are highly exothermic or involve unstable intermediates. The reduced reactor volumes also allow for higher temperatures and

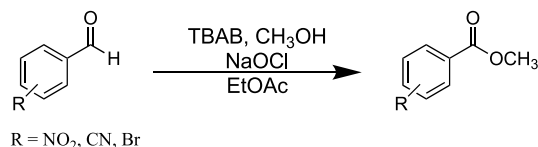
pressures to be used without compromising safety. Given these advantages, continuous flow techniques are now widely regarded as an important green chemistry tool.^{47,48} Two experiments that introduce flow techniques in the organic chemistry undergraduate laboratory are described here. These experiments allow students to compare the “batch” synthesis (using round-bottom flasks as reaction vessels) and “continuous flow” synthesis in which pumps are used to move fluids into a reaction tube (microreactor) and again to collect the product at the end.

One advantage of this technology that is of relevance to an undergraduate laboratory is that the shorter reaction times typical in a flow method allow the experiment to be completed in a 3–4 h laboratory period. It can also be carried out on a smaller scale in the teaching laboratory, thus generating less waste.

Aldehyde Oxidation Using Bleach

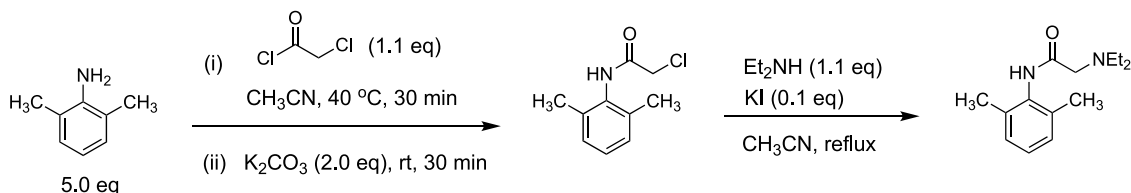
Collins and co-workers reported an experiment that involves the oxidative conversion of aldehydes to methyl esters using bleach and a phase transfer catalyst, tetra-*n*-butyl ammonium bromide (TBAB), in methanol (Scheme 24).⁴⁹

Scheme 24. Oxidative Conversion of Aldehydes to Methyl Esters Using Bleach and a Phase Transfer Catalyst



The authors reported that the regular batch synthesis required 2 h for completion, while in the continuous flow set up, the reaction was over in 5–15 min and the product was obtained in higher yields without the need for chromatographic purification. This reduces the amount of solvent needed for the overall reaction. One of the complications reported is the crystallization of 4-nitrobenzaldehyde in the reaction tube, but this was prevented by using a warmer water bath. Overall this experiment allows students to carry out reactions using continuous flow set up and discuss the importance of concepts such as mixing in a reaction, green oxidation, and phase transfer catalysts. The continuous flow set up uses two commercially available syringe pumps. One of the syringes contains the phase transfer catalyst,

Scheme 23. Synthesis of Lidocaine

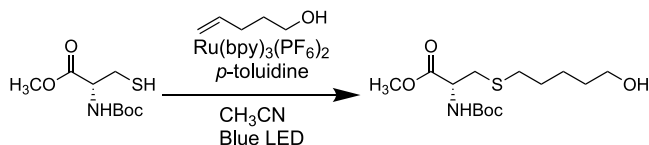


methanol, and ethyl acetate. The second syringe is used to pump the solution of aqueous bleach; with a T-mixer both materials are combined to be pumped into the reactor coil tube that is placed in a water bath (25–30 °C) and the product is collected in a round-bottom flask.

Photochemical Thiol–Ene Reaction

Collins and co-workers described the use of visible light instead of harmful UV light to promote thiol–ene alkylation of a protected cysteine derivative. They compare results from both a batch synthesis and a continuous flow synthesis (Scheme 25).⁵⁰

Scheme 25. Thiol–Ene Alkylation Promoted by Blue LED



The reaction was carried out using blue LED in acetonitrile and required tris(2,2'-bipyridine) ruthenium(II) hexafluorophosphate as a catalyst and *p*-toluidine as the cocatalyst. The authors reported shorter reaction times for photochemical continuous flow (2 min) versus conventional methods (2 h) with similar yields and conversions. Purification of the products by chromatography was carried out using ethyl acetate/hexanes as a solvent in both cases. Again, a switch from hexane to heptane would make the experiment greener by improving safety. One of the drawbacks in the continuous flow method is the use of excess acetonitrile (LD₅₀ = 617 mg/kg, mouse, oral) as a solvent to allow all the reaction mixture to flow through the reaction tube. Overall this experiment exposes students to a reaction that requires a catalyst and a cocatalyst, in addition to the relatively uncommon use of visible light as promoter in a photochemical reaction.

MICROWAVE-ASSISTED REACTIONS

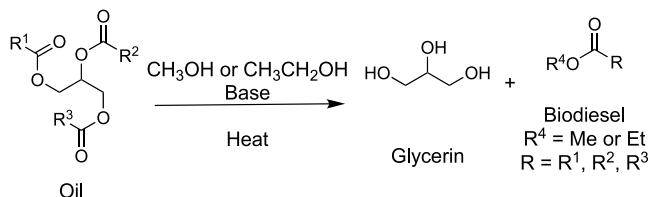
Microwave ovens are energy-efficient heating devices that reduce reaction times by transferring energy directly to the reactants. Reactions carried out in a microwave oven often give product(s) in higher yields, with greater selectivity and reduced byproduct(s) formation. Initially kitchen microwave ovens were used to develop methodologies that could be applied in the teaching laboratories; as mentioned before, the use of domestic (kitchen) microwave ovens has been discouraged for safety reasons. Microwave synthesis reactors have been developed to allow better control of the reaction conditions, improve safety, and increase reproducibility of the experiments. Two types of microwave ovens can be used in research and teaching laboratories: *Monomode microwave ovens* allow for one reaction at a time, providing even heating and ensuring reproducibility. *Multimode microwave ovens* allow for multiple reactions to be run simultaneously; to ensure homogeneous heating, multimode microwave ovens require a carousel.

Microwave-assisted organic synthesis has been applied to a variety of reactions such as esterification, saponification, nucleophilic and electrophilic aromatic substitution reactions, condensation reactions, coupling reactions, Diels–Alder reactions etc., not only in research laboratories but also in the teaching laboratories.

Esterification

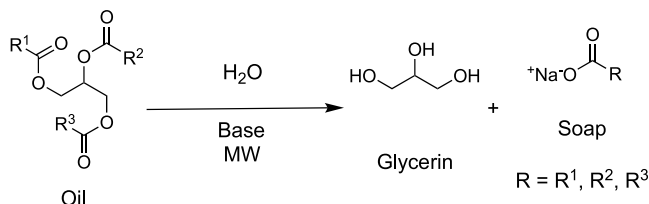
A microwave-assisted transesterification for the preparation of biodiesel from plant oil has been reported and compared with conventional heating methods (Scheme 26).^{51,52}

Scheme 26. Preparation of Biodiesel via Transesterification Promoted by Conventional Heating



Miller and Leadbeater⁵¹ reported similar yields for both conventional and microwave methods, while Pohl and co-workers⁵² reported higher yields for the microwave-assisted biodiesel synthesis. Products were analyzed by IR and NMR spectroscopy, and by polarimetry. Triglycerides from plant oils are chiral molecules since the ester chains are different. Pohl's transesterification and saponification reactions are followed by analysis of the product using polarimetry (Scheme 27) since the glycerol byproduct is achiral.

Scheme 27. Preparation of Biodiesel via a MW Assisted Transesterification



MW-Assisted Diels–Alder Reaction

One of the reactions covered in the organic curriculum that better illustrates the concept of atom economy is the Diels–Alder reaction. A microwave-assisted D–A reaction using absolute ethanol (a better microwave absorber than ethyl acetate) as the solvent reduced the reaction time from 2.5 h (conventional reflux) to 11 min (Scheme 28).^{53,54}

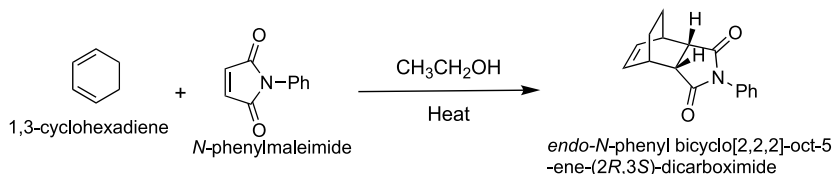
The use of microwave ovens is also desirable from a safety perspective since ethanol and 1,3-cyclohexadiene are highly flammable materials. Reagents and products should be handled with care since they are irritants and toxic [*N*-phenylmaleimide (LD₅₀ = 128–148.3 mg/kg, rat male–female, oral)].

Wittig Salt Synthesis

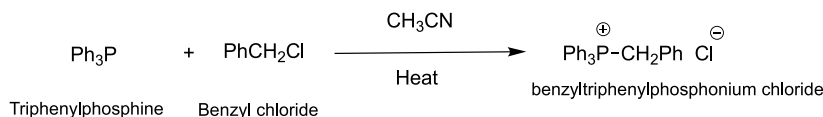
The synthesis of benzyltriphenylphosphonium chloride salt (Wittig salt) using a microwave oven in acetonitrile as a solvent reduced the reaction heating time from 1.5 h (reflux in xylenes) to 6.5 min with yields ranging from 90–100% (Scheme 29).^{53,54}

The reaction using xylenes (mixture of *o*-, *m*-, and *p*-isomers, (LD₅₀ = 5,000–6,602 mg/kg)) provided lower yields. Use of the slightly more toxic solvent acetonitrile (LD₅₀ = 617 mg/kg, mouse, oral) improved the efficiency because it is a better microwave energy absorber. Such differences can be used to illustrate to students how improving one parameter (yield) to improve a reaction can lead to a compromise in another aspect (safety). Xylenes are still used during the purification step of the Wittig salt. All reactants and products are toxic, or irritants,

Scheme 28. MW-Assisted Diels–Alder Reaction in Ethanol



Scheme 29. Synthesis of a Wittig Salt Using a MW Oven

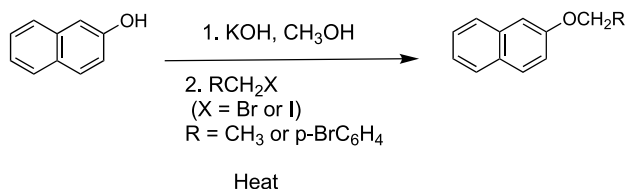


triphenylphosphine ($\text{LD}_{50} = 700 \text{ mg/kg}$, rat, oral), benzyl chloride ($\text{LD}_{50} = 560 \text{ mg/kg}$, rat, oral); the product, benzyltriphenylphosphonium chloride ($\text{LD}_{50} = 43 \text{ mg/kg}$, rat, oral) is highly toxic. Benzyl chloride is a lachrymator and should be handled with particular care.

Williamson Ether Synthesis

The formation of unsymmetrical ethers via an $\text{S}_{\text{N}}2$ reaction normally gives low yields and requires 1.5 h or more of reflux under conventional heating conditions while the microwave assisted reaction takes place within 10 min. The drawback of this reaction is the low yield of the product (30%). All the reagents are fairly toxic, 2-naphthol ($\text{LD}_{50} = 1,960 \text{ mg/kg}$, oral, rat), ethyl iodide ($\text{LD}_{50} = 330 \text{ mg/kg}$, oral, rat), *p*-bromobenzyl bromide ($\text{LD}_{50} = 1,960 \text{ mg/kg}$, oral, rat), thus making the experiment less suitable for use in an undergraduate laboratory (Scheme 30).^{53,54}

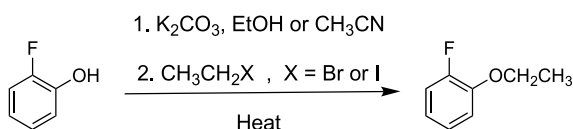
Scheme 30. Williamson Ether Synthesis of Naphthyl Ethers Using Conventional Heating and MW Irradiation



Baar also reported an inquiry-based, research-oriented experiment that illustrates the Williamson ether synthesis. This experiment allowed students to optimize the procedure using various microwave heating conditions, solvents, and concentration of reagents. Products were analyzed using gas chromatography. Reactions take between 1.5–3 min of heating, allowing ample time for two reactions to be completed during one typical laboratory period (Scheme 31).⁵⁵

The drawback of the experiment is the use of rather toxic 2-fluorophenol ($\text{LD}_{50} = 330 \text{ mg/kg}$, oral, rat) and ethyl bromide ($\text{LD}_{50} = 1,350 \text{ mg/kg}$, oral, rat), a suspected carcinogen.

Scheme 31. Williamson Ether Synthesis Using 2-Fluorophenol



E2 Dehydrohalogenation to Form Diphenylacetylene

Baar and co-workers also reported the formation of alkynes starting from alkenes via a multistep synthesis that involves the bromination of *E*-stilbene and then the bimolecular elimination reaction of the product, 1*R*,2*S*-dibromo-1,2-diphenylethane to form diphenylacetylene (Scheme 32).⁵⁴

These dehydrohalogenations require high temperatures and the use of a high boiling solvent, such as polyethylene glycol. The use of heating baths at 200 °C is a hazard, and the use of a high boiling solvent makes the purification of the product more difficult. Baar and co-workers demonstrated the use of microwave-assisted organic synthesis to achieve high temperatures under safer conditions and using ethanol as a solvent, thus reducing reaction times and obtaining good yields.

Fischer Esterification

Fischer esterification, a reaction often performed in the organic chemistry laboratory, requires fairly long reflux times. Baar and co-workers adapted a microwave-assisted Fischer esterification as an inquiry-based experiment in which students identify the products and starting materials based on NMR analysis of product (Scheme 33).⁵⁴

The use of both monomode and multimode microwave ovens provided good yields of the various esters (70–97%). Total heating time using the monomode oven was 2 min; for multimode ovens the time was 25–30 min. One of the drawbacks of the procedure presented in this experiment is the use of dichloromethane (carcinogen, $\text{LD}_{50} = 2,000 \text{ mg/kg}$) for purification of the products.

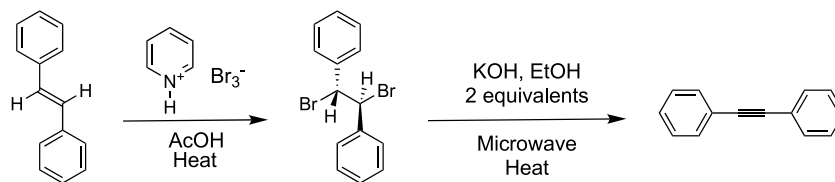
Nucleophilic Aromatic Substitution

Latimer and Wiebe compared the nucleophilic aromatic substitution reactions of 2,4-dinitrobenzene with one of the following nucleophiles: potassium thiocyanate, ethyl amine, or diethylamine.⁵⁶ The microwave-assisted substitution took place in 5 min with heating and used ethanol–water as the solvent system. Yields for these reactions improved by 1.3–2.0 times compared to the conventional heating method (1 h heating) using toluene as solvent (Scheme 34).

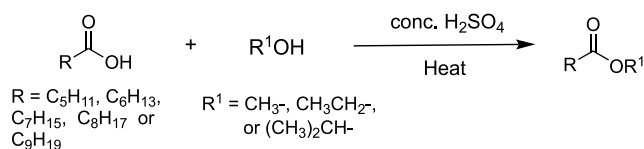
Students were asked to compare the two procedures using green chemistry metrics such as atom economy and environmental factor (E-factor).

Synthesis of a Diarylpyrazole

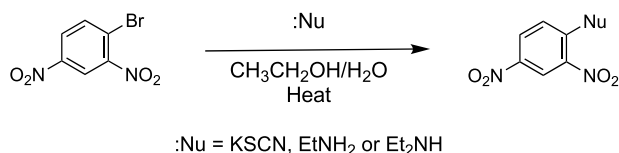
The synthesis of 1,5-diarylpyrazole (Celecoxib), a COX-2 inhibitor nonsteroidal anti-inflammatory drug, was adapted for the organic and medicinal chemistry laboratories by Anderluh using microwave-assisted organic synthesis (MAOS). The synthesis involves a Claisen condensation followed by formation

Scheme 32. Two-Step Synthesis of Diphenylacetylene from *E*-Stilbene Using MW Irradiation

Scheme 33. Fisher Esterification Catalyzed by Concentrated Sulfuric Acid



Scheme 34. Microwave Assisted Nucleophilic Aromatic Substitution Reaction



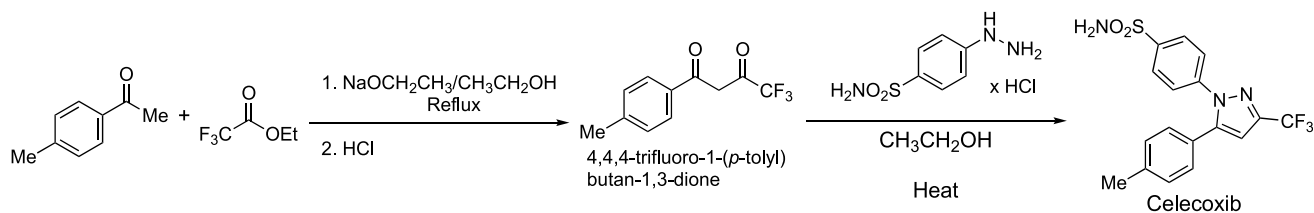
of a diarylpyrazole. Higher yields were obtained with the microwave-assisted method compared to one involving conventional heating (Scheme 35).⁵⁷

The regioselectivity of the diarylpyrazole condensation was also investigated by Anderluh and co-workers by ¹H NMR analysis of the crude product (Scheme 36). As previously reported by Reddy (reflux condition 95% overall yield, 99.31% Celecoxib and 0.5% 1,3-diarylpyrazole),⁵⁸ the use of higher temperatures in the microwave-assisted organic synthesis improved the yields of Celecoxib relative to 1,3-diarylpyrazole, which was formed only in trace amounts. (Scheme 36).⁵⁷

OXIDATION–REDUCTION REACTIONS

A significant part of the undergraduate organic chemistry curriculum is devoted to discussion of oxidation and reduction reactions. These reactions illustrate to students the important concept of interconversion of functional groups. Alcohols can be oxidized to aldehydes, ketones, or carboxylic acids and their derivatives. In the reverse reaction, carbonyl compounds and acid derivatives can be reduced to alcohols. Oxidation reactions in particular employ harsh and corrosive reagents and hence considerable efforts have been directed to develop greener oxidation methods. Several laboratory experiments that illustrate environmentally friendly oxidation–reduction reactions have been developed for use in the undergraduate laboratory.

Scheme 35. Synthesis of a 1,5-Diarylpyrazole (Celecoxib)



Oxidation of Alcohols

Oxidation of alcohols is one of the most well studied reactions in organic chemistry. Many reagents have been used for this purpose (e.g., chromic acid, potassium permanganate, Dess–Martin periodinane, pyridinium chlorochromate, and DMSO coupled with activating agents). In all cases the oxidizing agents raise safety concerns as they are typically corrosive and/or toxic, and often generate byproducts that are hazardous. Mohan and co-workers reported a discovery-oriented experiment using swimming pool chlorine (NaOCl) as an oxidizing agent. Students determined the selectivity of the oxidation using a mixture of a 1° and 2° alcohols (Scheme 37).⁵⁹

Oxidation of 2-heptanol occurred faster, while the concomitant slower oxidation of 1-hexanol led to formation of a dimeric ester as a byproduct, albeit in a low yield. A drawback is the use of the highly flammable diethyl ether for the extraction of the crude products. 2-Heptanol was chosen over 2-hexanol since the product of oxidation of the latter, 2-hexanone, is a neurotoxin.⁶⁰

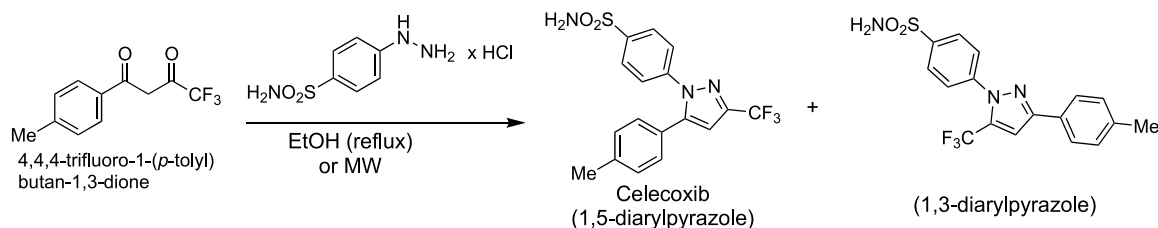
Vitaku and Christie reported an oxidation reaction using bleach and a biphasic system (ethyl acetate/water) using sodium bisulfate for the oxidation of (1*S*)-borneol. The (1*S*)-camphor product is purified by sublimation and analyzed by NMR spectroscopy (Scheme 38).⁶¹

Vitaku and Christie report the formation of aldehydes from benzylic alcohols unlike the alkyl esters observed by Mohan and co-workers⁵⁹ when NaOCl was used to oxidize 1-heptanol. It should, however, be noted that the dimeric products were formed in acetic acid.

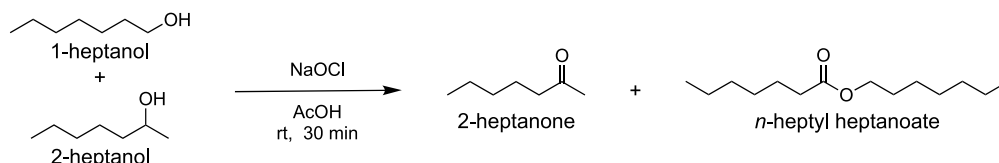
Oxidation of benzylic alcohols to benzaldehydes was reported by Stahl and co-workers using air as an oxidant and a copper/TEMPO catalyst (Scheme 39).⁶² Using air as an oxidant is especially attractive in the industry. (TEMPO = (2,2,6,6-tetramethyl-piperidin-1-yl)oxyl, bpy = 2,2'-bipyridine, and NMI = *N*-methyl imidazole). The initial species formed in this catalytic cycle is shown in Figure 1.

This experiment was designed as a discovery-oriented experiment in which students determine the identity of the starting material based on the analysis of the NMR spectra and EI-MS analysis of the product. Drawing catalytic cycles for the oxidation is a new and useful concept for most organic chemistry students who are typically more familiar with arrow pushing mechanisms.

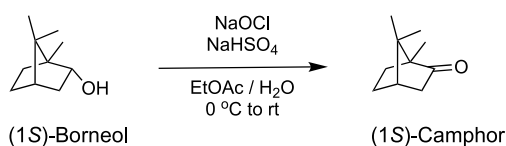
Scheme 36. Regioselectivity of the Diarylpyrazole Condensation



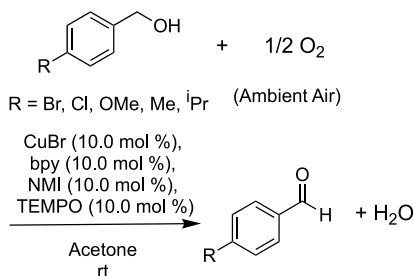
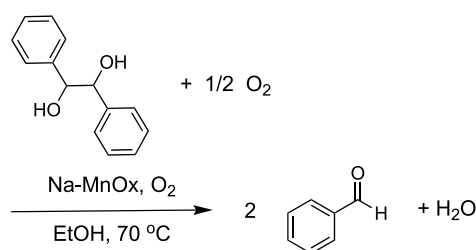
Scheme 37. Selective Oxidation of a 2° Alcohol Using Swimming Pool Chlorine



Scheme 38. Oxidation of (1S)-Borneol Using Bleach in a Biphasic System



Scheme 39. Oxidation of Benzylic Alcohols Using Air and TEMPO

Scheme 40. Oxidative Cleavage of *meso*-Hydrobenzoin Using a Heterogenous Catalyst

(H_2O_2). Readers are directed to the Supporting Information section of ref 63 for further details.

Dragojlovic and Chandrasekhar reported a green method for aromatization of 4,5-dimethyl-1,4-cyclohexadiene-1,2-dicarboxylic acid via bromination using in situ generated bromine, thus avoiding molecular bromine, which is highly corrosive and difficult to handle, and yielded 4,5-dimethyl phthalic acid (Scheme 41).⁶⁴ The reaction is a bromination followed by elimination to give the benzene ring. The intermediate bromo intermediate is shown in Figure 2.

They used a molecular modeling exercise to allow students to propose a mechanism for the formation of the aromatic product. Generation of bromine in situ should be done carefully as H_2O_2 (30%) is a strong oxidant and corrosive.

Bucholtz and co-workers reported a “benign by design” synthesis of a dihydropyridine. The use of a multicomponent Hantzsch synthesis followed by oxidative aromatization that uses an iron catalyst with hydrogen peroxide provided the desired pyridine compound. Water as well as acetic acid/water (50:50 by volume) were used as solvents for the reactions (Schemes 42 and 43).⁶⁵ Formaldehyde was used as a 37% solution in water.

The drawback of this experiment is the use of formaldehyde, which is carcinogenic. In addition dichloromethane is used as an extraction solvent ($\text{LD}_{50} > 2,000 \text{ mg/kg, oral, rat}$).

Hydrogenation

While hydrogenation reactions are typically atom economical, they often require high temperatures and pressures. O'Connor and co-workers used a commercially available catalyst, SiliaCat Pd(0), and a hydrogen balloon for the hydrogenation of alkenes and alkynes under mild conditions (Scheme 44).⁶⁶

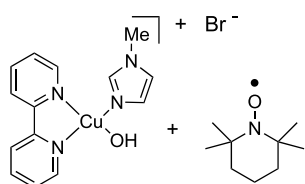


Figure 1. Copper/TEMPO catalyst system.

Oxidative Cleavage

Anastas and co-workers reported the oxidative cleavage of *meso*-hydrobenzoin to benzaldehyde using a heterogeneous catalyst (Na-MnOx) and oxygen in ethanol (Scheme 40).⁶³

Flame sources should be avoided during this experiment as students would be working with oxygen gas balloons and ethanol. Removal of the catalyst at the end of the experiment by filtration makes the purification of the product easy. Products are analyzed by TLC and GC. Students are asked to consider green metric concepts such as atom economy and E-factor for this reaction and compare it to oxidative cleavage using NaIO_4 (Malaprade) and Pb(OAc)_4 (Criegee) oxidations. Na-MnOx catalyst was prepared from manganese(II) chloride tetrahydrate ($\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$) by the addition of 30% hydrogen peroxide

Scheme 41. Aromatization of 4,5-Dimethyl-1,4-cyclohexadiene-1,2-dicarboxylic Acid via a Bromination–Elimination Sequence

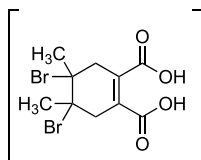
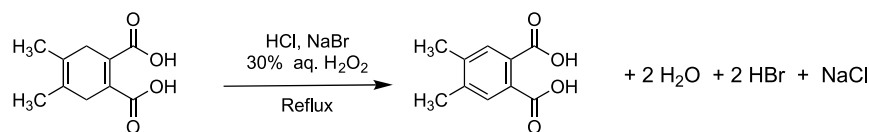


Figure 2. Bromo intermediate.

The ability to reuse the solvent and catalyst, the room temperature conditions, and the fact that no byproducts form, are some of the green chemistry principles that form part of the student discussion. Students are also encouraged to consider green metrics (atom economy and E-factor). Care must be taken to avoid flames and contact with air due to the use of H_2 gas. An attractive feature of the experiment is the use of 2-methyltetrahydrofuran, which is much safer than THF, being less prone to peroxide formation. 2-MethylTHF is obtained from sugars via furfural and this fact can be used to illustrate that chemicals can be obtained from renewable biosources and not just nonrenewable petroleum-based resources.⁶⁷

Reduction of an Aldehyde to an Alcohol

Wicht and co-workers used poly(methylhydro)siloxane (PMHS) with fluoride as a catalyst for the reduction of citronellal to citronellol (Scheme 45).⁶⁸

This reaction avoids the use of more common yet difficult to handle reducing agents such as LiAlH_4 and NaBH_4 .

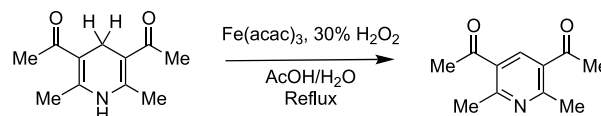
Wolff–Kishner Reduction

The use of solar irradiation as an alternative heat source has been implemented into a laboratory experiment by Swartling and co-workers in the Wolff–Kishner reduction of aryl ketones (Scheme 46).⁶⁹

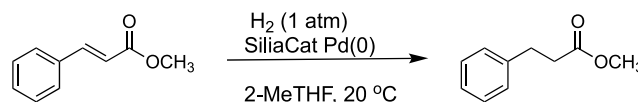
They compared the Wolff–Kishner reduction using conventional electrical heating with renewable solar heating. In order to make the reaction conditions safer, all reactions were carried out using propylene glycol ($\text{LD}_{50} = 20,000 \text{ mg/kg}$, oral, rat) instead of ethylene glycol. One limitation of a solar reactor is that there must be enough sunlight for it to work.

Solvent-Free Reactions

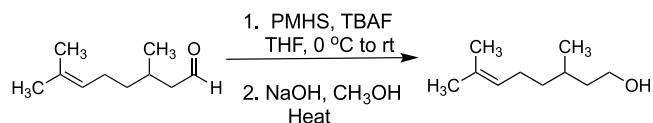
In order to decrease waste in the laboratory and industry, the use of solvent-free reactions is often encouraged. Dicks has cogently summarized in a review some solvent-free reactions developed for the organic chemistry laboratory curriculum.⁷⁰ Examples reported include Wittig reaction, crossed-aldol condensation, Friedel–Crafts reaction, and imine formation. When one of the reagents is a liquid, it is used as both reagent and solvent while other solvents are normally added during workup of the reactions or purification purposes. Caution must be exercised in

Scheme 43. Oxidative Aromatization of Dihydropyridine Using $\text{Fe}(\text{acac})_3$ 

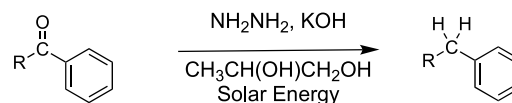
Scheme 44. Hydrogenation of Methylcinnamate Using SiliaCat Pd(0) as a Catalyst



Scheme 45. Reduction of Citronellal to Citronellol



Scheme 46. Wolff–Kishner Reduction of Aryl Ketones Using Solar Energy



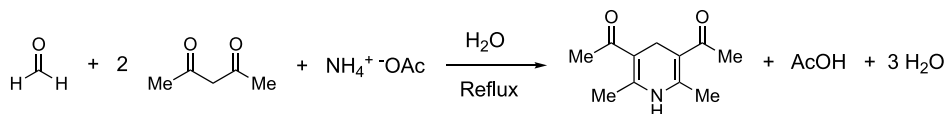
R = Me, Et, Pr, ⁱPr, Ph

reading manuscripts titled “solvent-free” as often one reagent is used in such an excess that this reagent effectively also serves as solvent!

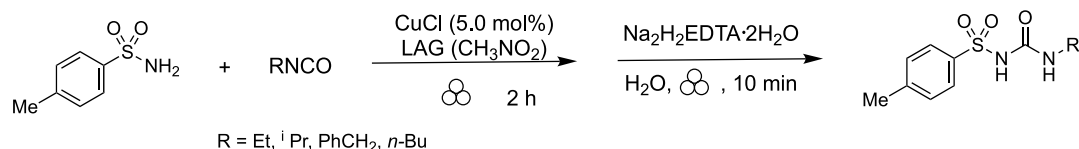
In the synthesis of active pharmaceutical ingredients, the use of mechanical synthesis (mechanochemical) such as grinding, milling, stretching, or shearing is on the rise. Mechanochemical reactions are considered “benign by design” because they can be done under solvent-free conditions. Coalcino and co-workers designed a solvent-free synthesis of Tolbutamide, a first-generation sulfonylurea-based oral antihypoglycemic drug.⁷¹ The reaction of an alkyl isocyanate with *p*-toluenesulfonamide using two types of mills—Mixer Mill (MM) and a Planetary Mill (PM)—provide the corresponding sulfonyl ureas in good yields. The reactions were carried out under solvent-free conditions or with liquid-assisted grinding (LAG) (Scheme 47).

LAG (also known as solvent-drop grinding) uses small amounts of a liquid (μL) to improve reactivity.^{72,73} In this case, a small amount of a nonhygroscopic, high boiling aprotic solvent such as nitromethane was used. Other solvents suitable for this

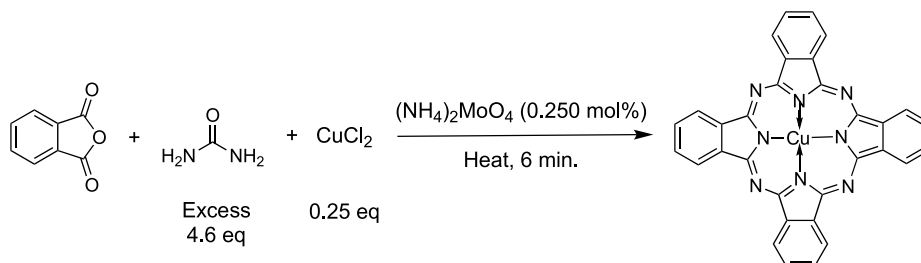
Scheme 42. Synthesis of Dihydropyridine via a Multicomponent Hantzsch Synthesis



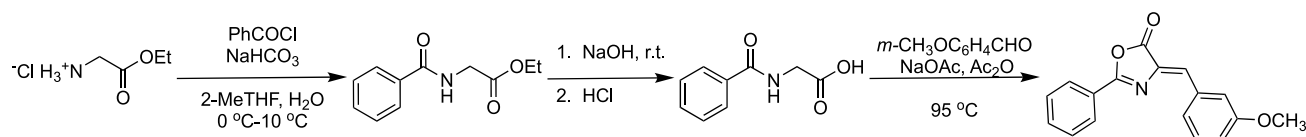
Scheme 47. Solvent-Free Synthesis of Tolbutamide



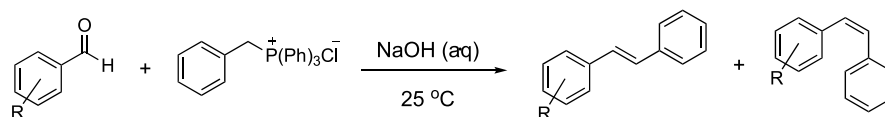
Scheme 48. Solvent-Free One-Pot Synthesis of Copper(II) Phthalocyanine



Scheme 49. Multistep Synthesis of an Azlactone



Scheme 50. Synthesis of (E) and (Z) Stilbene under Aqueous Conditions via a Wittig Reaction



purpose are acetone, acetonitrile, and ethyl methyl ketone. Purification of the product was done by filtration of the sulfonyl urea after removal of the Cu catalyst by complexation with disodium dihydrogen ethylenediamine tetraacetate dihydrate ($\text{Na}_2\text{H}_2\text{EDTA}\cdot 2\text{H}_2\text{O}$). The drawback of this experiment is the use of fairly toxic reagents: ethyl isocyanate ($\text{LD}_{50} = 56 \text{ mg/kg}$, intravenous, mouse), isopropyl isocyanate, and butyl isocyanate ($\text{LD}_{50} = 360 \text{ mg/kg}$, oral, rat). Benzyl isocyanate is toxic, flammable, and a lachrymator. Cu(I)Cl ($\text{LD}_{50} = 336 \text{ mg/kg}$, oral, rat) is an irritant and toxic.

Several solvent-free reactions use energy-efficient heating sources such as microwave-assisted synthesis. Sharma and co-workers reported a one-pot synthesis of copper(II) phthalocyanine. Copper chloride, phthalic anhydride, and urea in the presence of ammonium molybdate as a catalyst were heated using a microwave (Scheme 48).⁷⁴

The solid reagents and catalyst were homogenized using a mortar and pestle, and then heated using a microwave. The safety risk is the formation of ammonia gas, which is toxic. The reaction produces the phthalocyanine in 90% yield with this procedure, and the unreacted starting materials were washed out with diluted acid and base solutions.

Formation of Azlactone Derivatives

An open-ended experiment in which students are asked to prepare an azlactone target molecule was devised by Dicks and co-workers.⁷⁵ Students need to consider green chemistry concepts to achieve the synthesis, starting from glycine ethyl ester hydrochloride. A representative proposed synthesis is shown here (Scheme 49).

Students are asked to write a research proposal for the synthesis of the azlactone assigned to them. This experiment allows students to adapt a conventional synthesis and incorporate green chemistry principles to improve the reaction conditions. An example is replacing chloroform ($\text{LD}_{50} = 908 \text{ mg/kg}$, oral, rat; toxic if inhaled and suspected carcinogen) with 2-methyltetrahydrofuran ($\text{LD}_{50} > 300 \text{ mg/kg}$, oral, rat; flammable and irritant to skin and eyes), which is obtained from a renewable resource. Such toxicity data should serve as a reminder that LD_{50} values alone should not serve as a metric for determining the more suitable solvent. Also, triethylamine was replaced with a nontoxic base, sodium bicarbonate ($\text{LD}_{50} = 4.2 \text{ g/kg}$, oral, rat). Students are encouraged to discuss the green aspects of their synthesis, point out the drawbacks, and propose ways to make the reaction greener.

Aqueous Wittig Reaction

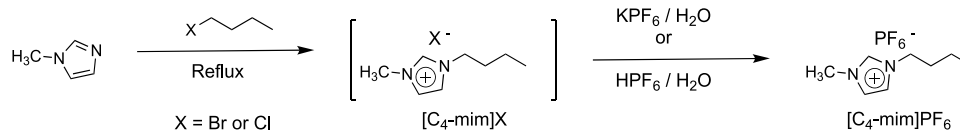
Morsh and co-workers reported a Wittig reaction using aromatic aldehydes under aqueous conditions to form (E) and (Z) alkenes at room temperature (Scheme 50).⁷⁶

While a traditional Wittig reaction is typically carried out in the undesirable solvent, hexane, the green approach in this Wittig reaction includes the use of water as a solvent to generate the ylide. Product purification by filtration and recrystallization reduces the amount of waste normally generated by liquid-liquid extraction and chromatography.

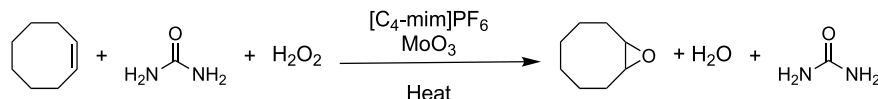
Ionic Liquids

Ionic liquids have become popular in the past two decades as alternative solvents for organic synthesis. They are low melting salts that are liquids at or near room temperature, and contain an

Scheme 51. Synthesis of an Ionic Liquid, 1-Butyl-3-methylimidazolium Hexafluorophosphate



Scheme 52. Epoxidation of Cyclooctene in an Ionic Liquid



organic cation combined with organic or inorganic anions. They have low coordinating abilities and are highly solvating over a wide range of temperatures. Although once touted as green solvents, they have limitations that must be carefully considered when adapting them as solvents. For example, many reports in the literature mention ionic liquids as alternatives to organic solvents, and yet ironically in these reactions the product is often isolated by extraction with generous amounts of organic solvents! Ionic liquids are typically recovered by extraction with other volatile organic solvents and can be reused relatively easily.⁷⁷

The synthesis of the ionic liquid 1-butyl-3-methylimidazolium hexafluorophosphate, $[C_4\text{-mim}]\text{PF}_6$, has been introduced as an undergraduate chemistry laboratory experiment by two groups: Dzyuba and co-workers,⁷⁸ and Galindo and co-workers (Scheme 51).⁷⁹

The synthesis of $[C_4\text{-mim}]\text{PF}_6$ has been achieved by an anion exchange of 1-butyl-3-methylimidazolium halide, $[C_4\text{-mim}]\text{X}$. The synthesis of $[C_4\text{-mim}]\text{X}$ is a good example of an S_N2 reaction. Even though ionic liquids are considered relatively safe solvents, many of the reagents used in the synthesis and purification are toxic to humans and harmful to the environment. 1-Bromobutane ($\text{LD}_{50} = 2.76 \text{ g/kg}$, oral, rat) and 1-chlorobutane ($\text{LD}_{50} = 2.67 \text{ g/kg}$, oral, rat) are toxic. 1-Methylimidazole ($\text{LD}_{50} = 1.14 \text{ g/kg}$, oral, rat) is combustible and toxic. Hexafluorophosphoric acid ($\text{LD}_{50} = 2.76 \text{ g/kg}$, oral, rat) is corrosive and toxic. Although still corrosive, potassium hexafluorophosphate (causes burns to skin and eyes) is a safer reagent for the anion exchange compared to HPF_6 .

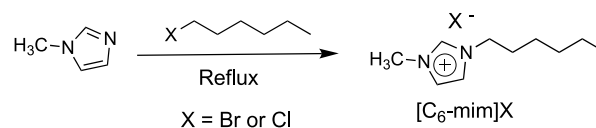
Purification of $[C_4\text{-mim}]\text{X}$ is done by washing with ethyl acetate, water, or dichloromethane. The use of dichloromethane by Dzyuba and co-workers⁷⁸ to reduce viscosity and to purify the final $[C_4\text{-mim}]\text{PF}_6$ partly defeats the purpose of using an ionic liquid. Water can also be used to reduce viscosity, though it can hydrolyze the PF_6 anion, especially at higher temperatures, to generate highly toxic HF and hence caution must be exercised in handling the ionic liquid.⁸⁰

Galindo and co-workers⁷⁹ demonstrated the use of the ionic liquid $[C_4\text{-mim}]\text{PF}_6$ as a solvent in the epoxidation of cyclooctene using a molybdenum catalyst with hydrogen peroxide as the oxidant (Scheme 52). They studied the effect of water on this oxidation reaction. Dry hydrogen peroxide (urea–hydrogen peroxide [UHP] adduct) and aqueous hydrogen peroxide were compared. Aqueous peroxide produced the 1,2-diol while the UHP reagent afforded the epoxide. They recovered and recycled the catalyst/ionic liquid ($[\text{MoO}_3]/[C_4\text{-mim}]\text{PF}_6$) solution mixture and tested its activity in a second catalytic oxidation reaction.⁷⁹

Stark and co-workers reported the synthesis of 1-hexyl-3-methylimidazolium halide ($[C_6\text{-mim}]\text{X}$) using several parame-

ters to better understand the green metrics needed to optimize these types of reactions (Scheme 53).⁸¹

Scheme 53. Synthesis of 1-Hexyl-3-methylimidazolium Halides



The study included the use of two alkylating agents, 1-bromo and 1-chlorohexane, as well as various sources of energy (conventional vs microwave), different heating temperatures, and different lengths of times. The factors analyzed include yield, atom economy, reaction mass efficiency, E-factor, energy efficiency (kg/kJ), price of chemicals (currency unit/kg), and toxicological data.

CONCLUSIONS AND PERSPECTIVE

There is an ever-increasing need for green and sustainable chemistry experiments for the undergraduate curriculum. However, restraint must be exercised when making claims about greenness of experimental procedures. While many claims are valid and backed up by data, exaggerated claims of greenness are not uncommon and should be discouraged. Recognizing that the term “green” refers more to a spectrum and not a magic number, authors may be well-advised to leave out the term “green” in titles of manuscripts, and instead discuss green aspects of the experiment in the text of the article.

There is also the ever-present gap between adoption of green chemistry principles in industries and academic laboratories, with industry obviously well ahead in the game. For example, while industries recognize the toxicity of *n*-hexane and have switched to *n*-heptane, academic laboratories worldwide are notorious for the large amounts of hexanes used for column chromatography to often isolate small amounts of a desired product. It is unclear what it will take to change this mind set. Perhaps principal investigators at large and famous laboratories should set an example by not just avoiding *n*-hexane but by also encouraging graduate students to find greener ways to their synthetic targets, and not just be driven by yield and quick routes to the target at the cost of compromising safety. Another example concerns oxidizing agents. While many nontoxic oxidizing agents have been adapted by industry, most standard organic chemistry textbooks in use would leave second-year chemistry students with the impression that the preferred oxidizing agents are based on carcinogenic chromium(VI) reagents!

Green chemistry principles are best inculcated at the undergraduate level and to that extent green chemistry experiments play a vital role. Most undergraduate laboratory experiments have focused on using safer reagents or on improving green metrics such as atom economy. More attention should be paid to what type of solvents are used, and to eliminating them when possible as solvents contribute to significant waste in a typical undergraduate laboratory.

AUTHOR INFORMATION

Corresponding Author

Ram S. Mohan — Department of Chemistry and Biochemistry, Illinois Wesleyan University, Bloomington, Illinois 61701, United States; orcid.org/0000-0003-4535-8324; Email: rmohan@iwu.edu

Author

Maria P. Mejia — Department of Chemistry and Biochemistry, Illinois Wesleyan University, Bloomington, Illinois 61701, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.jchemed.9b00753>

Notes

The authors declare no competing financial interest.

REFERENCES

- (1) (a) Anastas, P. T.; Kirchhoff, M. M. Origins, Current Status, and Future Challenges of Green Chemistry. *Acc. Chem. Res.* **2002**, *35* (9), 686–694. (b) Clark, J. H. Green Chemistry: Today (and Tomorrow). *Green Chem.* **2006**, *8* (1), 17–21. (c) Horvath, I. T.; Anastas, P. T. Innovations and Green Chemistry. *Chem. Rev.* **2007**, *107* (6), 2169–2173. (d) Anastas, P.; Eghbali, N. Green Chemistry: Principles and Practice. *Chem. Soc. Rev.* **2010**, *39* (1), 301–312.
- (2) United States Congress. H.R. 5931—Pollution Prevention Act of 1990. <https://www.congress.gov/bill/101st-congress/house-bill/5931> (accessed Feb 9, 2020).
- (3) Winterton, N. Twelve More Green Chemistry Principles. *Green Chem.* **2001**, *3* (6), G73–G75.
- (4) (a) *Green Organic Chemistry in Lecture and Laboratory*, 3rd ed.; Dicks, A. P., Ed.; CRC Press: New York, 2012. (b) *Green Organic Chemistry: Strategies, Tools and Laboratory Experiments*, 1st ed.; Doxsee, K. M., Hutchinson, J. E., Eds.; Brooks/Cole: Florence, KY, 2004. (c) *Green Chemistry Experiments in Undergraduate Laboratories*, 1st ed.; Fahey, J. T., Maelia, L. E., Eds.; American Chemical Society, Oxford University Press: Washington, D.C., 2016.
- (5) Ribeiro, M. G. T. C.; Costa, D. A.; Machado, A. A. S. C. Green Star: A Holistic Green Chemistry Metric for Evaluation of Teaching Laboratory Experiments. *Green Chem. Lett. Rev.* **2010**, *3* (2), 149–159.
- (6) Hodge, A.; Sterner, B. Canadian Centre for Occupational Health and Safety. *How Do I Compare One LD₅₀ Value to Another and What Does It Mean to Humans?* (Tables 1 and 2). <https://www.ccohs.ca/oshanswers/chemicals/ld50.html> (accessed Feb 9, 2020).
- (7) Lasker, G. A.; Simcox, N. J.; Mellor, K. E.; Mullins, M. L.; Nesmith, S. M.; van Bergen, S.; Anastas, P. T. Introducing Toxicology into the Undergraduate Chemistry Laboratory Using Safety Data Sheets and Screenshot Activities. *J. Chem. Educ.* **2019**, *96* (4), 720–724.
- (8) Duarte, R. C.; Ribeiro, M. G. T. C.; Machado, A. A. S. C. Reaction Scale and Green Chemistry: Microscale or Macroscale, Which Is Greener? *J. Chem. Educ.* **2017**, *94* (9), 1255–1264.
- (9) Hutchison, J. E. Systems Thinking and Green Chemistry: Powerful Levers for Curricular Change and Adoption. *J. Chem. Educ.* **2019**, *96* (12), 2777–2783.
- (10) Jones-Wilson, T. M.; Burtch, E. A. A Green Starting Material for Electrophilic Aromatic Substitution for the Undergraduate Organic Laboratory. *J. Chem. Educ.* **2005**, *82* (4), 616–617.
- (11) Yadav, U.; Mande, H.; Ghalsasi, P. Nitration of Phenols Using Cu(NO₃)₂: Green Chemistry Laboratory Experiment. *J. Chem. Educ.* **2012**, *89* (2), 268–270.
- (12) (a) Chang, Y.-C. Neurotoxic effects of *n*-Hexane on the Human Central Nervous System: Evoked Potential Abnormalities in *n*-Hexane Polyneuropathy. *J. Neurol., Neurosurg. Psychiatry* **1987**, *50* (3), 269–274. (b) Takeuchi, Y.; Ono, Y.; Hisanaga, N.; Kitoh, J.; Suguira, Y. A Comparative Study on the Neurotoxicity of *n*-Pentane, *n*-Hexane, and *n*-Heptane in the Rat. *J. Neurol., Neurosurg. Psychiatry* **1987**, *50*, 269–274.
- (13) *Journal of Organic Chemistry Author Guidelines*. (See “Data Requirements” for guidelines on reporting use of a microwave device to conduct synthesis experiments.) https://publish.acs.org/publish/author_guidelines?coden=joceah (accessed Jan 8, 2020).
- (14) Dintzner, M. R.; Maresh, J. J.; Kinzie, C. R.; Arena, A. F.; Speltz, T. A Research-Based Undergraduate Organic Laboratory Project: Investigation of a One-Pot, Multicomponent, Environmentally Friendly Prins-Friedel–Crafts-Type Reaction. *J. Chem. Educ.* **2012**, *89* (2), 265–267.
- (15) Domling, A.; Wang, W.; Wang, K. Chemistry and Biology of Multicomponent Reactions. *Chem. Rev.* **2012**, *112* (6), 3083–3135.
- (16) Hopson, R.; Lee, P. Y. B.; Hess, K. M. 1-Dimensional Selective Nuclear Overhauser Effect NMR Spectroscopy To Characterize Products from a Two-Step Green Chemistry Synthesis. *J. Chem. Educ.* **2018**, *95* (4), 641–647.
- (17) Biswas, R.; Mukherjee, A. Introducing the Concept of Green Synthesis in the Undergraduate Laboratory: Two-Step Synthesis of 4-Bromoacetanilide from Aniline. *J. Chem. Educ.* **2017**, *94* (9), 1391–1394.
- (18) (a) Cave, G. W. V.; Raston, C. L. Green Chemistry Laboratory: Benign Synthesis of 4,6-Diphenyl[2,2′]bipyridine via Sequential Solventless Aldol and Michael Addition Reactions. *J. Chem. Educ.* **2005**, *82* (3), 468–469. (b) Bennett, G. D. A Green Enantioselective Aldol Condensation for the Undergraduate Organic Laboratory. *J. Chem. Educ.* **2006**, *83* (12), 1871–1872. (c) Pereira, J.; Afonso, C. A. M. Asymmetric Aldol Reaction Induced by Chiral Auxiliary. *J. Chem. Educ.* **2006**, *83* (9), 1333–1335. (d) Crouch, R. D.; Richardson, A.; Howard, J. L.; Harker, R. L.; Barker, K. H. The Aldol Addition and Condensation: The Effect of Conditions on Reaction Pathway. *J. Chem. Educ.* **2007**, *84* (3), 475–476.
- (19) Ter Meer Guardia, L. K.; Belli, A. J.; Molta, G. J.; Gordon, P.; Jaworek-Lopes, C. H. Green Crossed Aldol Condensation Reactions Using *trans*-Cinnamaldehyde. *Chem. Educ.* **2011**, *16*, 23–25.
- (20) Horta, J. E. Simple Microwave-Assisted Claisen and Dieckmann Condensation Experiments for the Undergraduate Organic Chemistry Laboratory. *J. Chem. Educ.* **2011**, *88* (7), 1014–1015.
- (21) Stacey, J. M.; Dicks, A. P.; Goodwin, A. A.; Rush, B. M.; Nigam, M. Green Carbonyl Condensation Reactions Demonstrating Solvent and Organocatalyst Recyclability. *J. Chem. Educ.* **2013**, *90* (8), 1067–1070.
- (22) Chen, J.; Spear, S. K.; Huddleston, J. G.; Rogers, R. D. Polyethylene Glycol and Solutions of Polyethylene Glycol as Green Reaction Media. *Green Chem.* **2005**, *7*, 64–82.
- (23) Kumar, H. M. S.; Subbareddy, B. V.; Anjaneyulu, S.; Yadav, J. S. Non Solvent Reaction: Ammonium Acetate Catalyzed Highly Convenient Preparation of *Trans*-Cinnamic Acids. *Synth. Commun.* **1998**, *28*, 3811–3815.
- (24) Keuseman, K. J.; Morrow, N. C. A “Green” Approach to Synthesis of *trans*-4-Methoxycinnamic Acid in the Undergraduate Teaching Laboratory. *Chem. Educ.* **2014**, *19*, 347–350.
- (25) Mullins, J. J.; Prusinowski, A. F. Microwave-Promoted Synthesis of a Carbocyclic Curcuminoid: An Organic Chemistry Laboratory Experiment. *J. Chem. Educ.* **2019**, *96* (3), 606–609.
- (26) Egorova, K. S.; Ananikov, V. P. Toxicity of Metal Compounds: Knowledge and Myths. *Organometallics* **2017**, *36* (21), 4071–4090.

- (27) Pantess, D. A.; Rich, C. V. Aqueous Suzuki Reactions: A Greener Approach to Transition Metal-Mediated Aryl Couplings in the Organic Instructional Laboratory. *Chem. Educ.* **2009**, *14*, 258–260.
- (28) Costa, N. E.; Pelotte, A. L.; Simard, J. M.; Syvinski, C. A.; Deveau, A. M. Discovering Green, Aqueous Suzuki Coupling Reactions: Synthesis of Ethyl (4-Phenylphenyl)acetate, a Biaryl with Anti-Arthritic Potential. *J. Chem. Educ.* **2012**, *89* (8), 1064–1067.
- (29) Huddleston, J. G.; Visser, A. E.; Reichert, W. M.; Willauer, H. D.; Broker, G. A.; Rogers, R. D. Characterization and Comparison of Hydrophilic and Hydrophobic Room Temperature Ionic Liquids Incorporating the Imidazolium Cation. *Green Chem.* **2001**, *3* (4), 156–164.
- (30) Soares, P.; Fernandes, C.; Chavarria, D.; Borges, F. Microwave-Assisted Synthesis of 5-Phenyl-2-hydroxy-acetophenone Derivatives by a Green Suzuki Coupling Reaction. *J. Chem. Educ.* **2015**, *92* (3), 575–578.
- (31) Ramgren, S. D.; Hie, L.; Ye, Y.; Garg, N. K. Nickel-Catalyzed Suzuki–Miyaura Couplings in Green Solvents. *Org. Lett.* **2013**, *15* (15), 3950–3953.
- (32) Hie, L.; Chang, L. L.; Garg, N. K. Nickel-Catalyzed Suzuki–Miyaura Cross-Coupling in a Green Alcohol Solvent for an Undergraduate Organic Chemistry Laboratory. *J. Chem. Educ.* **2015**, *92* (3), 571–574.
- (33) Palesch, J. J.; Gilles, B. C.; Chycota, J.; Haj, M. K.; Fahnhorst, G. W.; Wissinger, J. E. Iodination of Vanillin and Subsequent Suzuki–Miyaura Coupling: Two-Step Synthetic Sequence Teaching Green Chemistry Principles. *Green Chem. Lett. Rev.* **2019**, *12* (2), 117–126.
- (34) Morra, B.; Dicks, A. P. Recent Progress in Green Undergraduate Laboratory Design. In *Green Chemistry Experiments in Undergraduate Laboratories*; Fahey, J. T., Maelia, L. E., Ed.; ACS Symposium Series; American Chemical Society: Washington, D.C., 2016; Vol. 1233, pp 26–27.
- (35) Ballard, E. C. Green Reductive Homocoupling of Bromobenzene. *J. Chem. Educ.* **2011**, *88* (8), 1148–1151.
- (36) (a) Nicolaou, K. C.; Snyder, S. A.; Montagnon, T.; Vassilikogiannakis, G. The Diels–Alder Reaction in Total Synthesis. *Angew. Chem., Int. Ed.* **2002**, *41* (10), 1668–1698. (b) Gregoritz, M.; Brandl, F. P. The Diels–Alder Reaction: A Powerful Tool for the Design of Drug Delivery Systems and Biomaterials. *Eur. J. Pharm. Biopharm.* **2015**, *97*, 438–453.
- (37) Otto, S.; Engberts, J. B. F. N. Diels–Alder Reactions in Water. *Pure Appl. Chem.* **2000**, *72* (7), 1365–1372.
- (38) Rugutt, J. K.; Pohl, T. L.; Grills, B. G.; Park, T.; Thanomsing, C.; Christian, N.; Rugutt, K. J.; Koech, J.; Obel, M. C.; Cheruiyot, G.; Chirchir, D.; Rop, R. An Undergraduate Research-based Organic Chemistry Lab Project on Guided Inquiry Green Synthesis of Novel Crystalline Diels–Alder Monomers for Ring Opening Metathesis Polymerization (ROMP). *Chem. Educ.* **2013**, *8*, 290–294.
- (39) Amin, S.; Barnes, A.; Buckner, C.; Jones, J.; Monroe, M.; Nurmomade, L.; Pinto, T.; Starkey, S.; Agee, B. M.; Crouse, D. J.; Swartling, D. J. Diels–Alder Reaction Using a Solar Irradiation Heat Source Designed for Undergraduate Organic Chemistry Laboratories. *J. Chem. Educ.* **2015**, *92* (4), 767–770.
- (40) Lehman, J. W. *Microscale Operational Organic Chemistry: A Problem-Solving Approach to the Laboratory Course*; Pearson Prentice Hall: Upper Saddle River, NJ, 2004.
- (41) Aslanian, R. Identification of an Unknown Diene in Eucalyptus Oil: A Modified Diels–Alder Experiment. *Chem. Educ.* **2017**, *22*, 73–76.
- (42) Bastin, L. D.; Nigam, M.; Martinus, S.; Maloney, J. E.; Benyack, L. L.; Gainer, B. Synthesis of Substituted N-Phenylmaleimides and Use in a Diels–Alder Reaction: A Green Multi-Step Synthesis for an Undergraduate Organic Chemistry Laboratory. *Green Chem. Lett. Rev.* **2019**, *12* (2), 127–135.
- (43) Goldstein, S. W. Biodiesel from Seeds: An Experiment for Organic Chemistry. *J. Chem. Educ.* **2014**, *91* (10), 1693–1696.
- (44) Constable, D. J. C.; Dunn, P. J.; Hayler, J. D.; Humphrey, G. R.; Leazer, J. L., Jr.; Linderman, R. J.; Lorenz, K.; Manley, J.; Pearlman, B. A.; Wells, A.; Zaks, A.; Zhang, T. Y. Key Green Chemistry Research Areas—A Perspective from Pharmaceutical Manufacturers. *Green Chem.* **2007**, *9* (5), 411–420.
- (45) Fennie, M. W.; Roth, J. M. Comparing Amide-Forming Reactions Using Green Chemistry Metrics in an Undergraduate Organic Laboratory. *J. Chem. Educ.* **2016**, *93* (10), 1788–1793.
- (46) Josephson, P.; Nykvist, V.; Qasim, W.; Blomkvist, B.; Diner, P. Student-Driven Development of Greener Chemistry in Undergraduate Teaching: Synthesis of Lidocaine Revisited. *J. Chem. Educ.* **2019**, *96* (7), 1389–1394.
- (47) Poechlauer, P.; Colberg, J.; Fisher, E.; Jansen, M.; et al. Pharmaceutical Roundtable Study Demonstrates the Value of Continuous Manufacturing in the Design of Greener Processes. *Org. Process Res. Dev.* **2013**, *17*, 1472–1478.
- (48) Gutmann, B.; Cantillo, D.; Kappe, C. O. Continuous-Flow Technology—A Tool for the Safe Manufacturing of Active Pharmaceutical Ingredients. *Angew. Chem., Int. Ed.* **2015**, *54*, 6688–6728.
- (49) Kairouz, V.; Collins, S. K. Continuous Flow Science in an Undergraduate Teaching Laboratory: Bleach-Mediated Oxidation in a Biphasic System. *J. Chem. Educ.* **2018**, *95*, 1069–1072.
- (50) Santandrea, J.; Kairouz, V.; Collins, S. K. Continuous Flow Science in an Undergraduate Teaching Laboratory: Photocatalytic Thiol–Ene Reaction Using Visible Light. *J. Chem. Educ.* **2018**, *95*, 1073–1077.
- (51) Miller, T. A.; Leadbeater, N. E. Microwave Assisted Synthesis of Biodiesel in an Undergraduate Organic Chemistry Laboratory Course. *Chem. Educ.* **2009**, *14*, 98–104.
- (52) Pohl, N. L. B.; Streff, J. M.; Brokman, S. Evaluating Sustainability: Soap versus Biodiesel Production from Plant Oils. *J. Chem. Educ.* **2012**, *89* (8), 1053–1056.
- (53) Baar, M. R.; Falcone, D.; Gordon, C. Microwave Enhanced Organic Synthesis for the Undergraduate Laboratory: Diels–Alder Cycloaddition, Wittig Reaction, and Williamson Ether Synthesis. *J. Chem. Educ.* **2010**, *87*, 84–86.
- (54) Baar, M. R.; Gammerding, W.; Leap, J.; Morales, E.; Shikora, J.; Weber, M. H. Pedagogical Comparison of Five Reactions Performed under Microwave Heating in Multi-Mode versus Mono-Mode Ovens: Diels–Alder Cycloaddition, Wittig Salt Formation, E2 Dehydrohalogenation to Form an Alkyne, Williamson Ether Synthesis, and Fischer Esterification. *J. Chem. Educ.* **2014**, *91*, 1720–1724.
- (55) Baar, M. R. Research Experience for the Organic Chemistry Laboratory: A Student-Centered Optimization of a Microwave-Enhanced Williamson Ether Synthesis and GC Analysis. *J. Chem. Educ.* **2018**, *95*, 1235–1237.
- (56) Latimer, D.; Wiebe, M. Greening the Organic Chemistry Laboratory: A Comparison of Microwave Assisted and Classical Nucleophilic Aromatic Substitution Reactions. *Green Chem. Lett. Rev.* **2015**, *8* (2), 39–42.
- (57) Anderluh, M. Synthesis of Celecoxib in the Undergraduate Organic Chemistry or Medicinal Chemistry Laboratories; Conventional vs. Microwave Heating. *Chem. Educ.* **2013**, *18* (2), 155–158.
- (58) Reddy, A. R.; Sampath, A.; Goverdhan, G.; Yakambaram, B.; Mukkanti, K.; Reddy, P. P. An Improved and Scalable Process for Celecoxib: A Selective Cyclooxygenase-2 Inhibitor. *Org. Process Res. Dev.* **2009**, *13*, 98–101.
- (59) Shadwick, S. R.; Mohan, R. S. The Discovery Oriented Approach to Organic Chemistry 2. Selectivity in Alcohol Oxidation. *J. Chem. Educ.* **1999**, *76*, 1121–1122.
- (60) Davenport, J. G.; Farrell, D. F.; Sumi, M. Giant Axonal Neuropathy Caused by Industrial Chemicals: Neurofilamentous Axonal Masses in Man. *Neurology* **1976**, *26* (10), 919–923.
- (61) Vitaku, E.; Christie, H. S. Improved Procedure for Bleach-Based Alcohol Oxidation in Undergraduate Laboratories. *J. Chem. Educ.* **2019**, *96*, 1042–1045.
- (62) Hill, N. J.; Hoover, J. M.; Stahl, S. S. Aerobic Alcohol Oxidation Using a Copper(I)/TEMPO Catalyst System: A Green, Catalytic Oxidation Reaction for the Undergraduate Organic Chemistry Laboratory. *J. Chem. Educ.* **2013**, *90*, 102–105.

- (63) Lam, C. H.; Escande, V.; Mellor, K. E.; Zimmerman, J. B.; Anastas, P. T. Teaching Atom Economy and E-Factor Concepts through a Green Laboratory Experiment: Aerobic Oxidative Cleavage of *meso*-Hydrobenzoin to Benzaldehyde Using a Heterogeneous Catalyst. *J. Chem. Educ.* **2019**, *96*, 761–765.
- (64) Chandrasekhar, C.; Dragojlovic, V. Reaction of Bromine with 4,5-Dimethyl-1,4-cyclohexadiene-1,2-dicarboxylic Acid: A Green Chemistry Puzzle for Organic Chemistry Students. *Green Chem. Lett. Rev.* **2010**, *3*, 39–47.
- (65) Bucholtz, K. M.; Hugdahl, J. D.; Kiefer, A. M.; Kiefer, K. M.; Santa Maria, A. M. A Green Multi-Component Synthesis in Organic Chemistry Laboratory: A Hantzsch Dihydropyridine Synthesis and Oxidation to the Corresponding Pyridine. *Chem. Educ.* **2012**, *17*, 125–127.
- (66) Pelfrey, C.; Ensel, S.; O'Connor, K. SiliaCat Pd(0), a New Green Hydrogenation Catalyst for the Undergraduate Organic Chemistry Laboratory. *Chem. Educ.* **2016**, *21*, 40–42.
- (67) Aycock, D. F. Solvent Applications of 2-Methyltetrahydrofuran in Organometallic and Biphasic Reactions. *Org. Process Res. Dev.* **2007**, *11*, 156–159.
- (68) O'Brien, K. E.; Wicht, D. K. A Greener Organic Chemistry Experiment: Reduction of Citronellal to Citronellol Using Poly-(methylhydro)siloxane. *Green Chem. Lett. Rev.* **2008**, *1* (3), 149–154.
- (69) Agee, B. M.; Mullins, G.; Biernacki, J. J.; Swartling, D. J. Wolf-Kishner Reduction Reactions Using a Solar Irradiation Heat Source and a Green Solvent System. *Green Chem. Lett. Rev.* **2014**, *7*, 383–392.
- (70) Dicks, A. P. Solvent-Free Reactivity in the Undergraduate Organic Laboratory. *Green Chem. Lett. Rev.* **2009**, *2*, 87–100.
- (71) Colacino, E.; Dayaker, G.; Morère, A.; Friscic, T. Introducing Students to Mechanochemistry via Environmentally Friendly Organic Synthesis Using a Solvent-Free Mechanochemical Preparation of the Antidiabetic Drug Tolbutamide. *J. Chem. Educ.* **2019**, *96*, 766–771.
- (72) Cheney, M. L.; McManus, G. J.; Perman, J. A.; Wang, Z.; Zaworotko, M. J. The Role of Cocrystals in Solid-State Synthesis: Cocrystal-Controlled Solid-State Synthesis of Imides. *Cryst. Growth Des.* **2007**, *7*, 616–617.
- (73) Cheney, M. L.; Zaworotko, M. J.; Beaton, S.; Singer, R. D. Cocrystal-Controlled Solid-State Synthesis. A Green Chemistry Experiment for Undergraduate Organic Chemistry. *J. Chem. Educ.* **2008**, *85*, 1649–1651.
- (74) Sharma, R. K.; Sharma, C.; Sidhwani, T. Solventless and One-Pot Synthesis of Cu(II) Phthalocyanine Complex: A Green Chemistry Experiment. *J. Chem. Educ.* **2011**, *88*, 86–87.
- (75) Edgar, L. J. G.; Koroluk, K. J.; Golmakani, M.; Dicks, A. P. Green Chemistry Decision-Making in an Upper-Level Undergraduate Organic Laboratory. *J. Chem. Educ.* **2014**, *91* (7), 1040–1043.
- (76) Morsch, L. A.; Deak, L.; Tiburzi, D.; Schuster, H.; Meyer, B. Green Aqueous Wittig Reaction: Teaching Green Chemistry in Organic Teaching Laboratories. *J. Chem. Educ.* **2014**, *91*, 611–614.
- (77) Chowdhury, S.; Scott, J. L.; Mohan, R. S. Reactivity of Ionic Liquids. *Tetrahedron* **2007**, *63*, 2363–2389.
- (78) Dzyuba, S. V.; Kollar, K. D.; Sabnis, S. S. Synthesis of Imidazolium Room-Temperature Ionic Liquids: Exploring Green Chemistry and Click Chemistry Paradigms in Undergraduate Organic Chemistry Laboratory. *J. Chem. Educ.* **2009**, *86*, 856–858.
- (79) Montilla, F.; Pastor, A.; Galindo, A. An Advanced Multi-Step Project in Green Chemistry: Using Ionic Liquids in MoO₃/H₂O₂ Catalyzed Epoxidation of *cis*-Cyclooctene. *Chem. Educ.* **2016**, *21*, 133–135.
- (80) Swatloski, R. P.; Holbrey, J. D.; Rogers, R. D. Ionic Liquids Are Not Always Green: Hydrolysis of 1-Butyl-3-methylimidazolium Hexafluorophosphate. *Green Chem.* **2003**, *5*, 361–363.
- (81) Stark, A.; Ott, D.; Kralisch, D.; Kreisel, G.; Ondruschka, B. Ionic Liquids and Green Chemistry: A Lab Experiment. *J. Chem. Educ.* **2010**, *87*, 196–201.